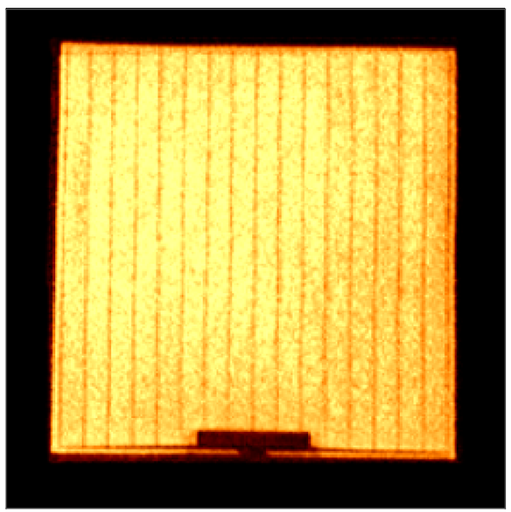
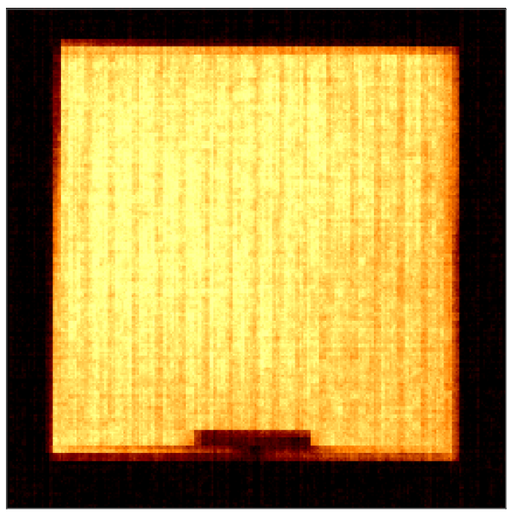
# Bad images for UTV | Reference_0021

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

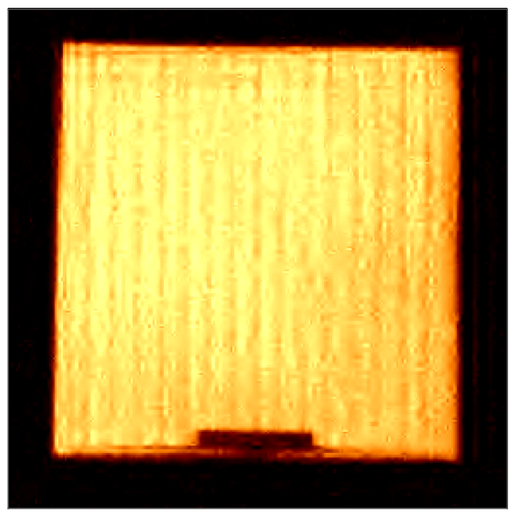
Ground truth
Ground truth



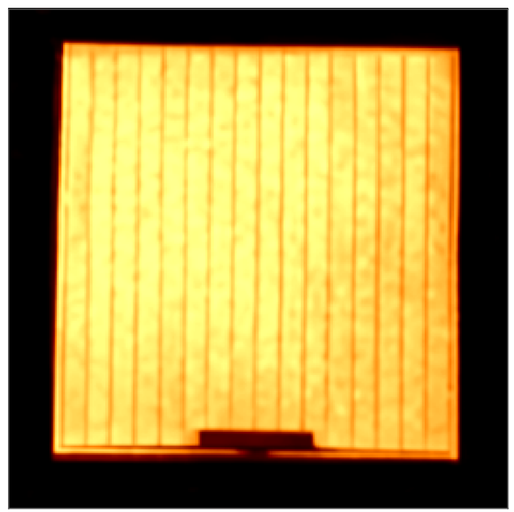
Zero-fill
PSNR 24.47 dB | SSIM 0.509
HaarPSI 0.564



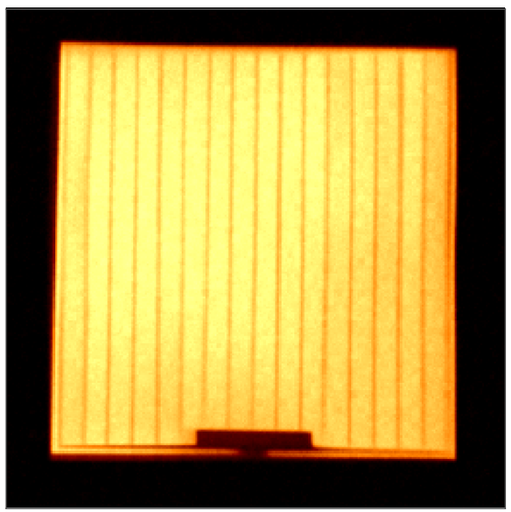
SPGL1-BPDN
PSNR 26.06 dB | SSIM 0.580
HaarPSI 0.591



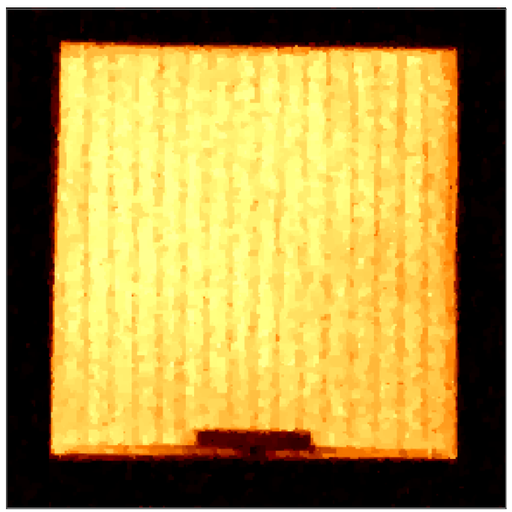
PnP-ADMM-tuned-5
PSNR 31.07 dB | SSIM 0.847
HaarPSI 0.822



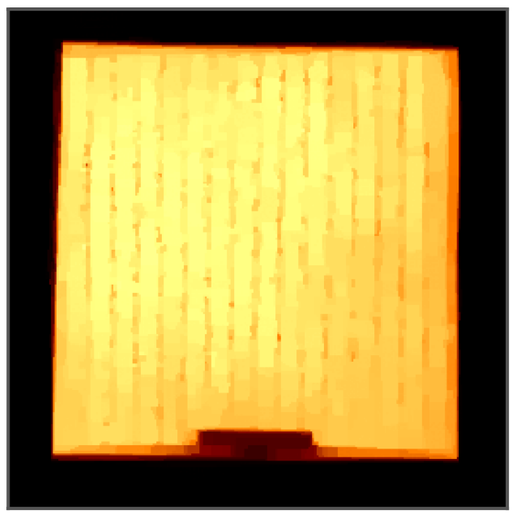
GS-PnP-tuned-5
PSNR 28.76 dB | SSIM 0.705
HaarPSI 0.755



Scalar TV
PSNR 27.48 dB | SSIM 0.642
HaarPSI 0.692



UTV C60
PSNR 27.83 dB | SSIM 0.695
HaarPSI 0.574






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

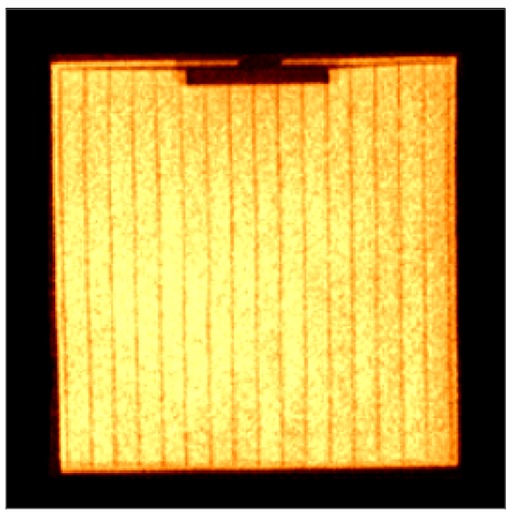
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

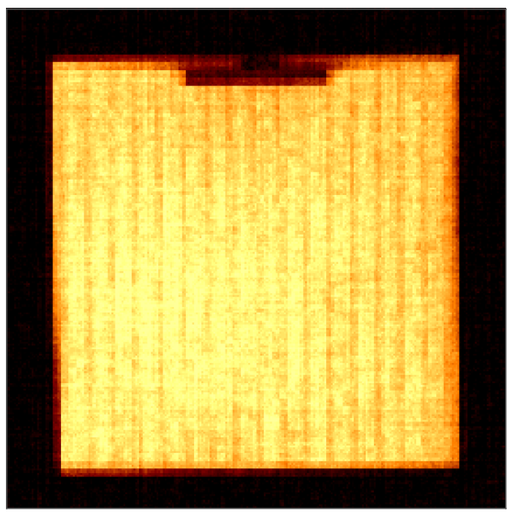
# Bad images for UTV | Reference_0066

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

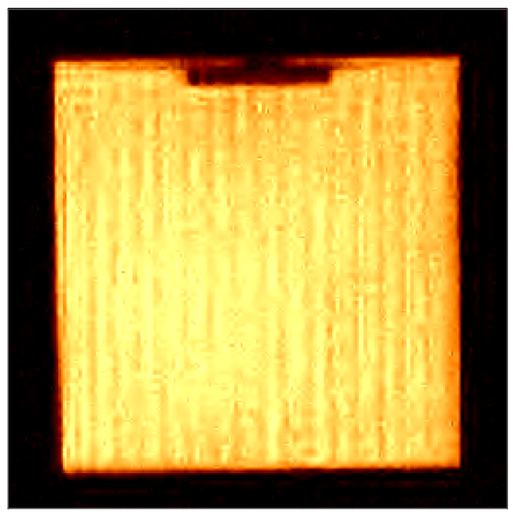
Ground truth
Ground truth



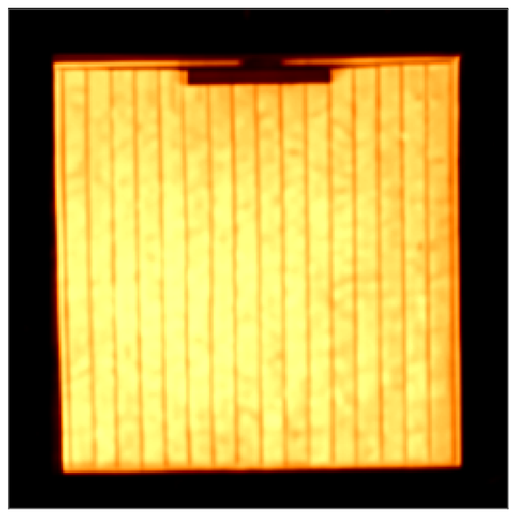
Zero-fill
PSNR 24.59 dB | SSIM 0.516
HaarPSI 0.544



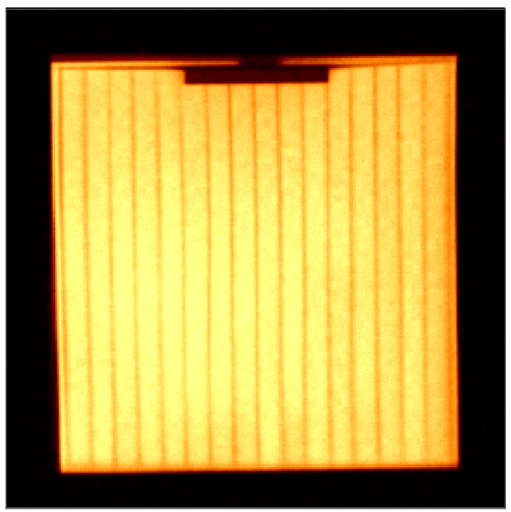
SPGL1-BPDN
PSNR 26.25 dB | SSIM 0.579
HaarPSI 0.599



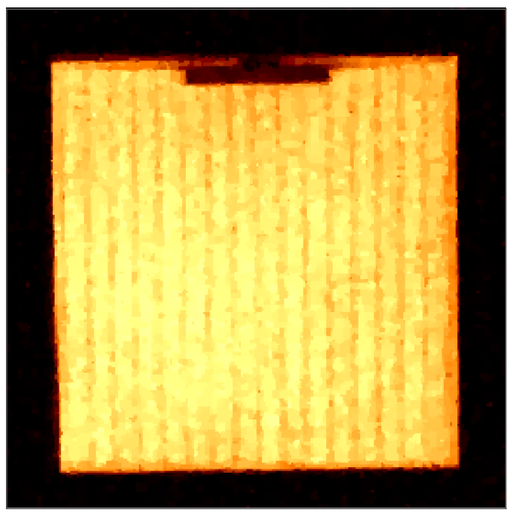
PnP-ADMM-tuned-5
PSNR 31.05 dB | SSIM 0.846
HaarPSI 0.814



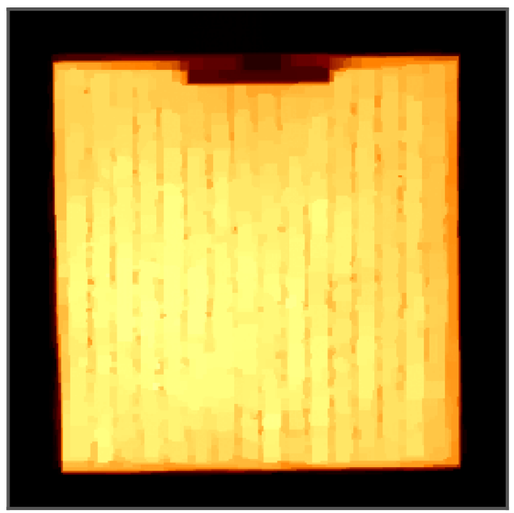
GS-PnP-tuned-5
PSNR 28.70 dB | SSIM 0.702
HaarPSI 0.748



Scalar TV
PSNR 27.54 dB | SSIM 0.650
HaarPSI 0.684



UTV C60
PSNR 27.67 dB | SSIM 0.688
HaarPSI 0.568






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

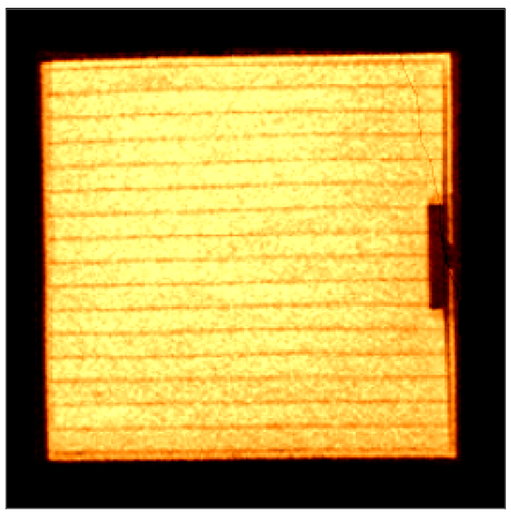
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

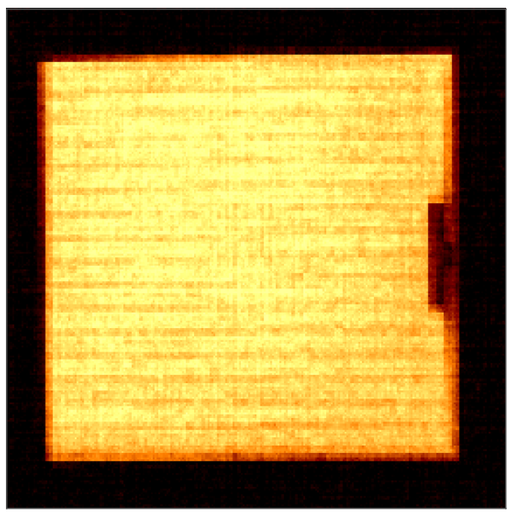
# Bad images for UTV | Reference_0188

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

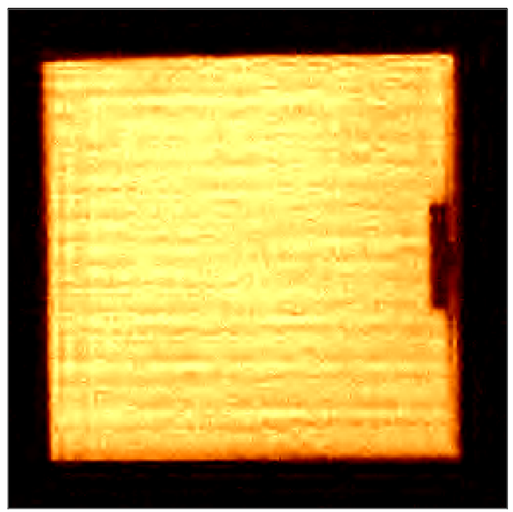
Ground truth
Ground truth



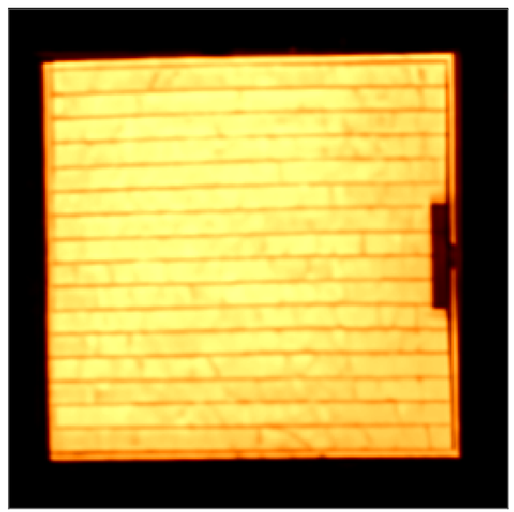
Zero-fill
PSNR 24.51 dB | SSIM 0.510
HaarPSI 0.565



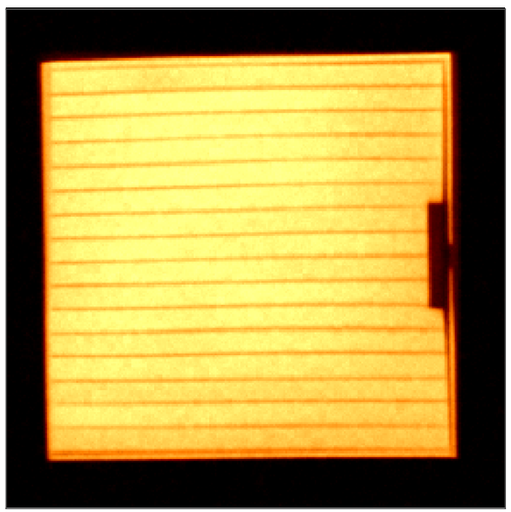
SPGL1-BPDN
PSNR 26.02 dB | SSIM 0.573
HaarPSI 0.606



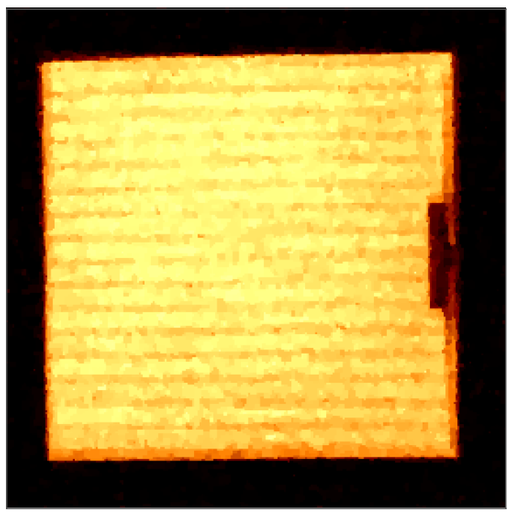
PnP-ADMM-tuned-5
PSNR 30.84 dB | SSIM 0.838
HaarPSI 0.818



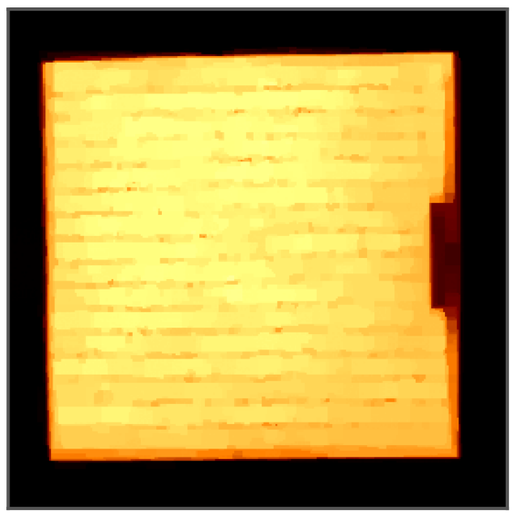
GS-PnP-tuned-5
PSNR 28.54 dB | SSIM 0.698
HaarPSI 0.744



Scalar TV
PSNR 27.35 dB | SSIM 0.643
HaarPSI 0.685



UTV C60
PSNR 27.74 dB | SSIM 0.700
HaarPSI 0.596






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

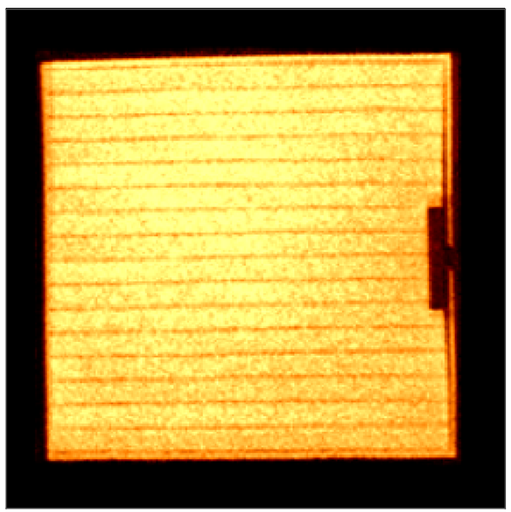
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

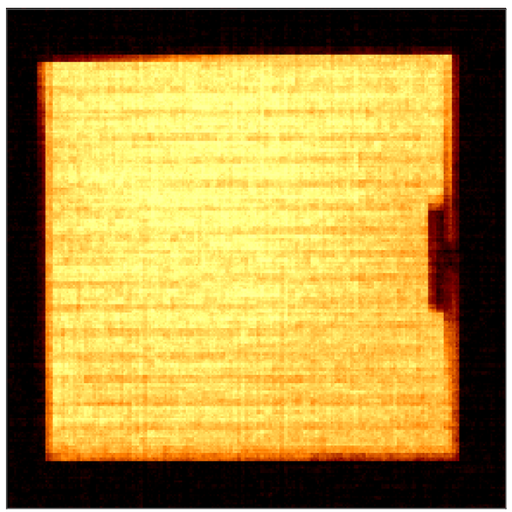
# Bad images for UTV | Reference_0199

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

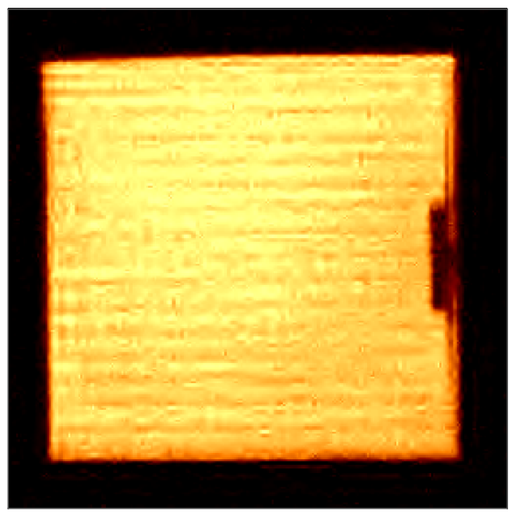
Ground truth
Ground truth



Zero-fill
PSNR 24.70 dB | SSIM 0.506
HaarPSI 0.578



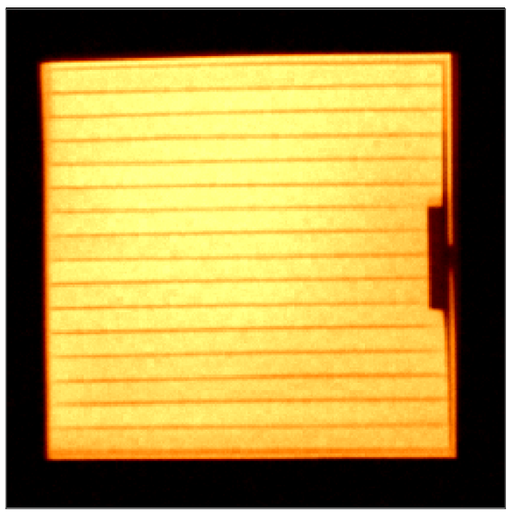
SPGL1-BPDN
PSNR 26.21 dB | SSIM 0.579
HaarPSI 0.598



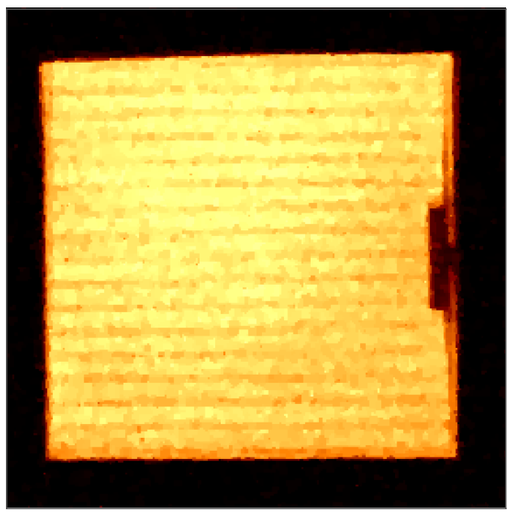
PnP-ADMM-tuned-5
PSNR 30.96 dB | SSIM 0.839
HaarPSI 0.819



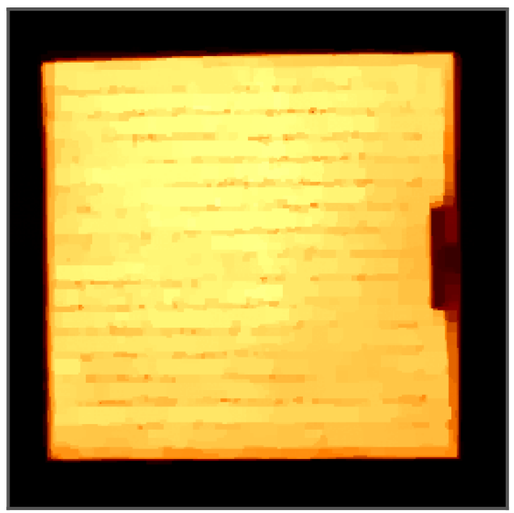
GS-PnP-tuned-5
PSNR 28.58 dB | SSIM 0.695
HaarPSI 0.747



Scalar TV
PSNR 27.56 dB | SSIM 0.640
HaarPSI 0.693



UTV C60
PSNR 27.77 dB | SSIM 0.688
HaarPSI 0.579






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

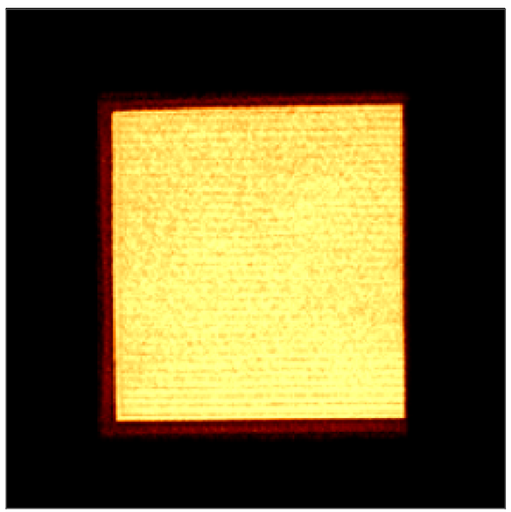
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

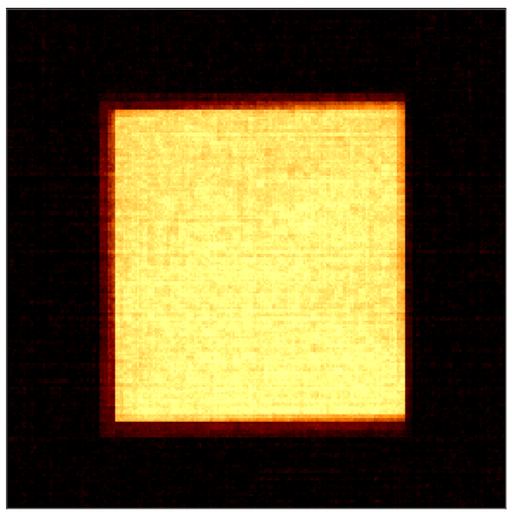
# Bad images for UTV | Reference_0260

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

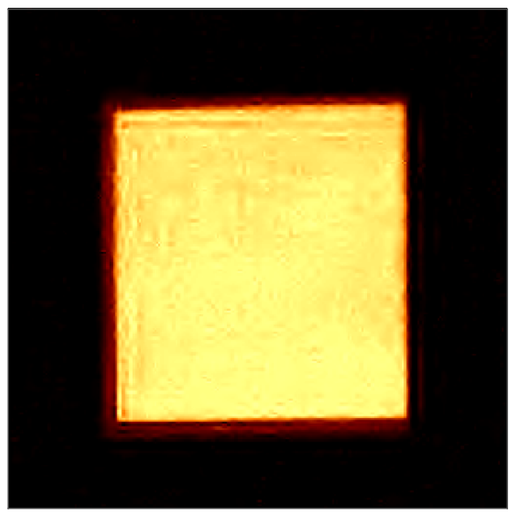
Ground truth
Ground truth



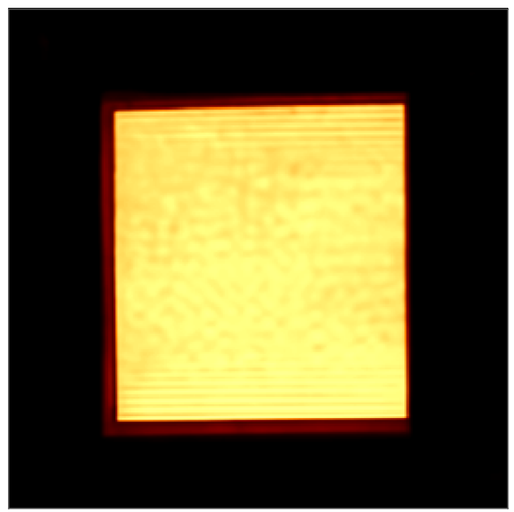
Zero-fill
PSNR 25.97 dB | SSIM 0.442
HaarPSI 0.532



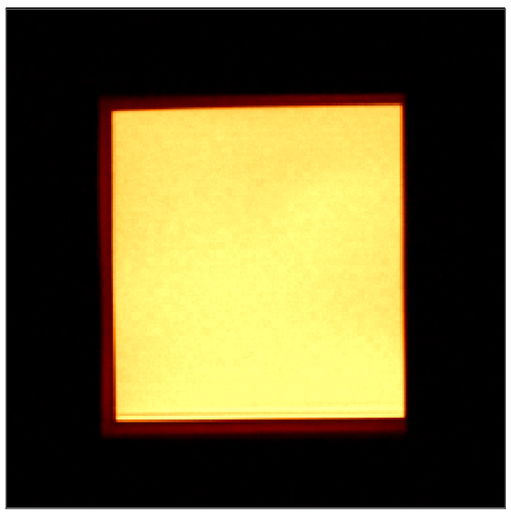
SPGL1-BPDN
PSNR 29.25 dB | SSIM 0.679
HaarPSI 0.618



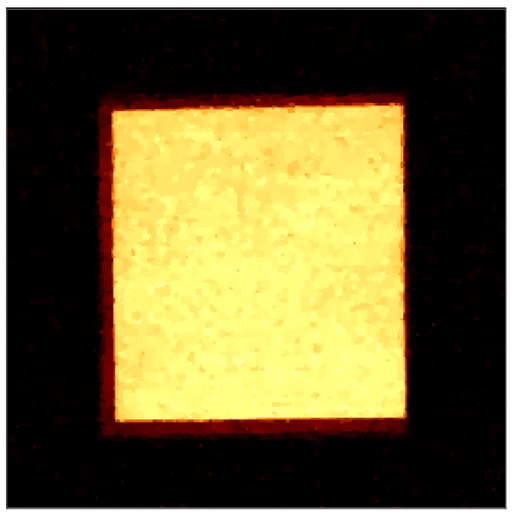
PnP-ADMM-tuned-5
PSNR 33.76 dB | SSIM 0.860
HaarPSI 0.825



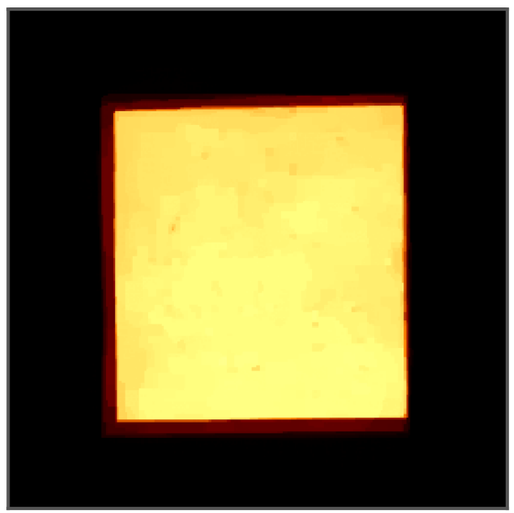
GS-PnP-tuned-5
PSNR 31.70 dB | SSIM 0.736
HaarPSI 0.733



Scalar TV
PSNR 30.61 dB | SSIM 0.638
HaarPSI 0.746



UTV C60
PSNR 32.38 dB | SSIM 0.791
HaarPSI 0.650






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

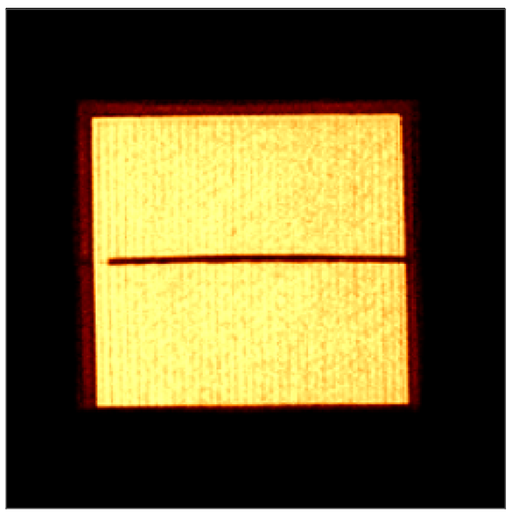
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Visual observation: all compared methods failed to recover the lines.

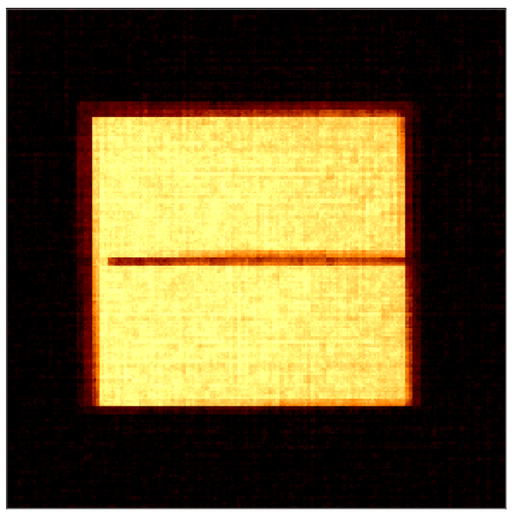
# Bad images for UTV | Reference_0357

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

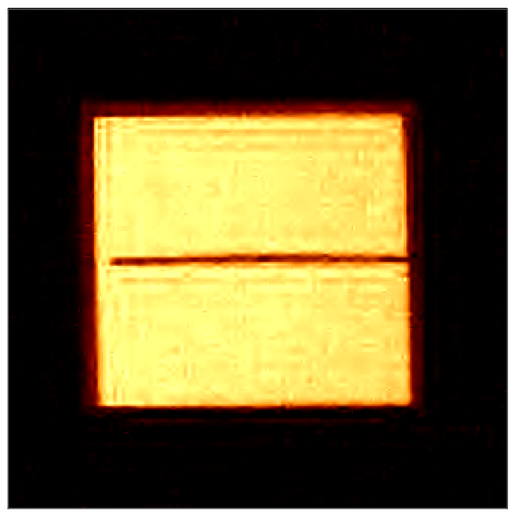
Ground truth
Ground truth



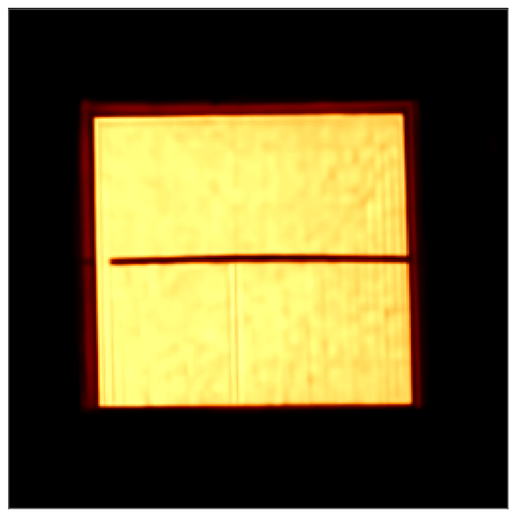
Zero-fill
PSNR 24.75 dB | SSIM 0.432
HaarPSI 0.526



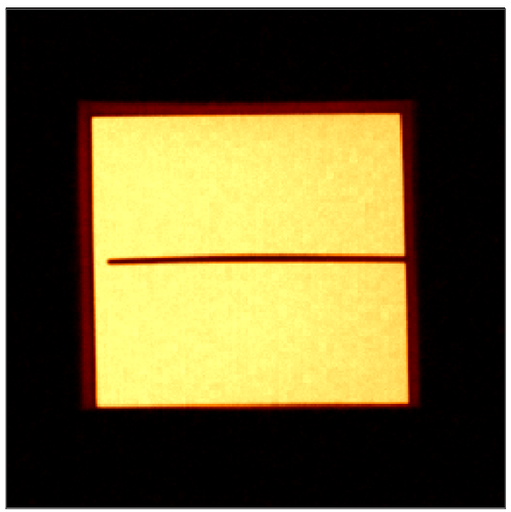
SPGL1-BPDN
PSNR 27.79 dB | SSIM 0.664
HaarPSI 0.585



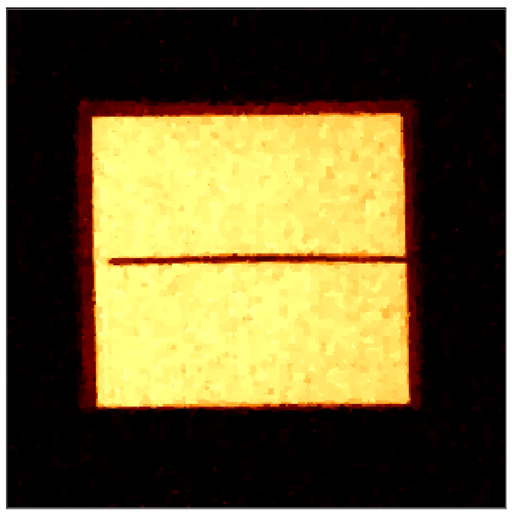
PnP-ADMM-tuned-5
PSNR 33.46 dB | SSIM 0.864
HaarPSI 0.806



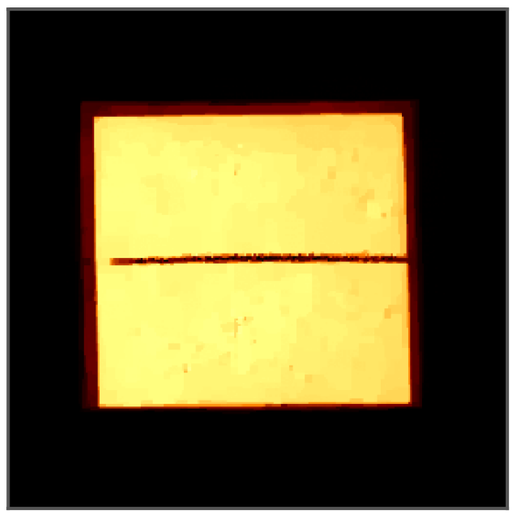
GS-PnP-tuned-5
PSNR 30.78 dB | SSIM 0.683
HaarPSI 0.715



Scalar TV
PSNR 30.04 dB | SSIM 0.649
HaarPSI 0.737



UTV C60
PSNR 31.50 dB | SSIM 0.802
HaarPSI 0.685






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

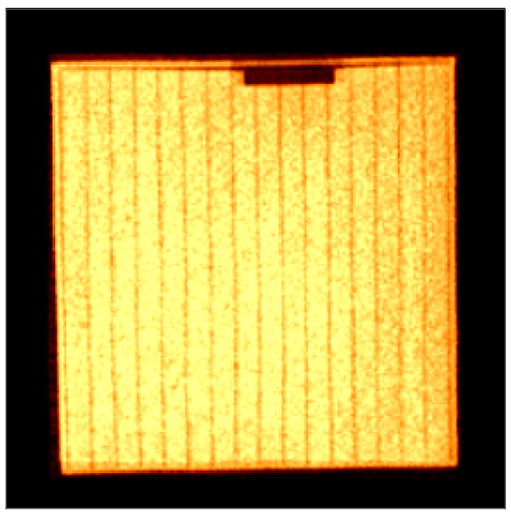
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

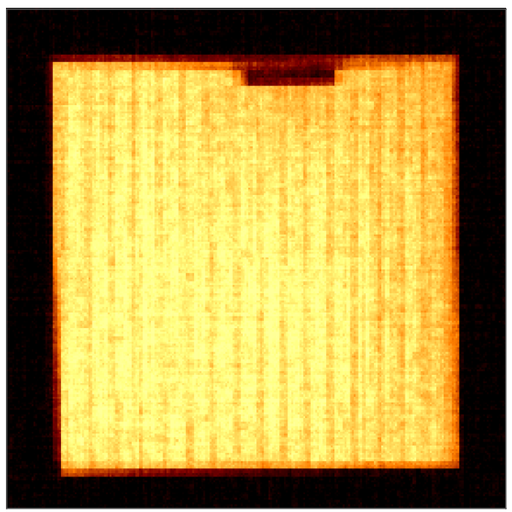
# Bad images for UTV | Reference_0727

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

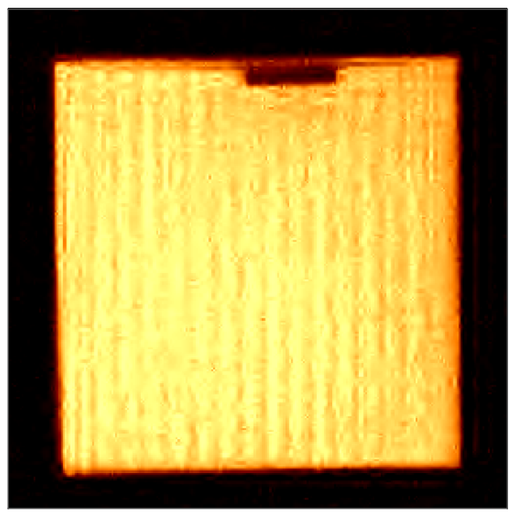
Ground truth
Ground truth



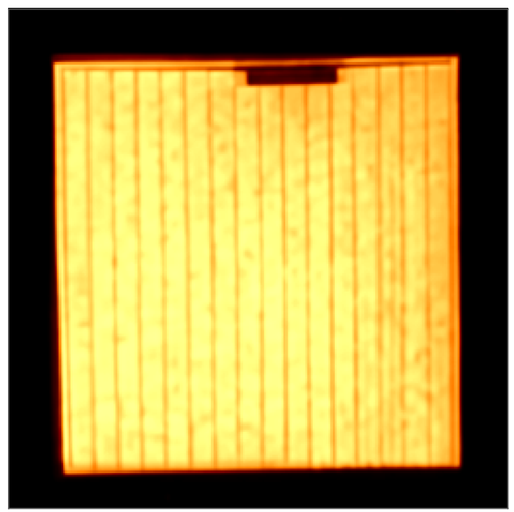
Zero-fill
PSNR 24.59 dB | SSIM 0.511
HaarPSI 0.574



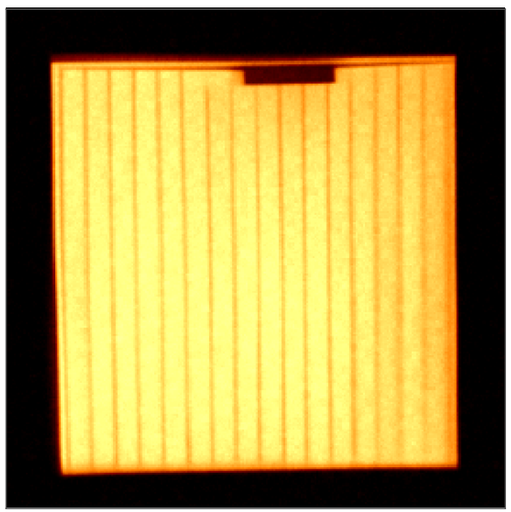
SPGL1-BPDN
PSNR 26.02 dB | SSIM 0.578
HaarPSI 0.607



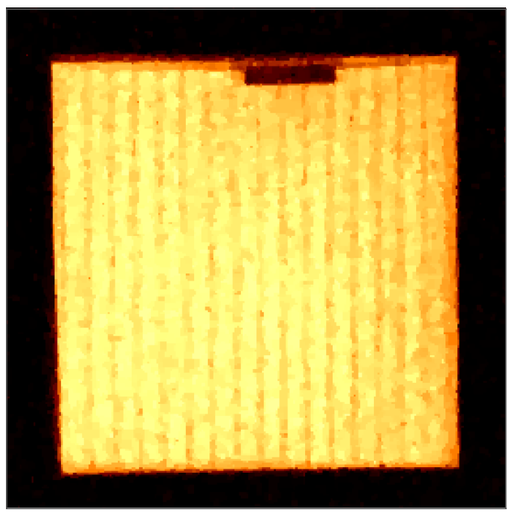
PnP-ADMM-tuned-5
PSNR 31.04 dB | SSIM 0.845
HaarPSI 0.828



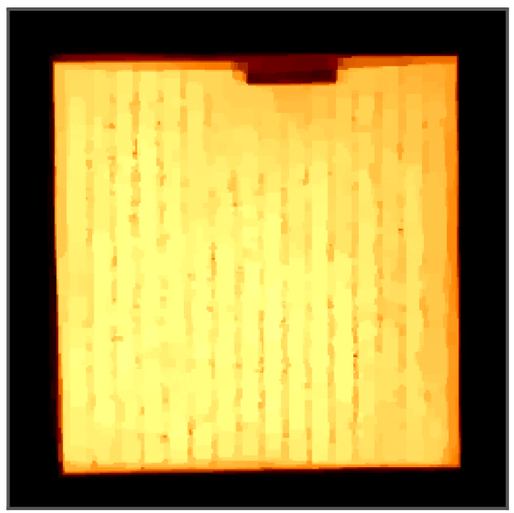
GS-PnP-tuned-5
PSNR 28.67 dB | SSIM 0.695
HaarPSI 0.752



Scalar TV
PSNR 27.44 dB | SSIM 0.647
HaarPSI 0.677



UTV C60
PSNR 27.83 dB | SSIM 0.698
HaarPSI 0.600






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

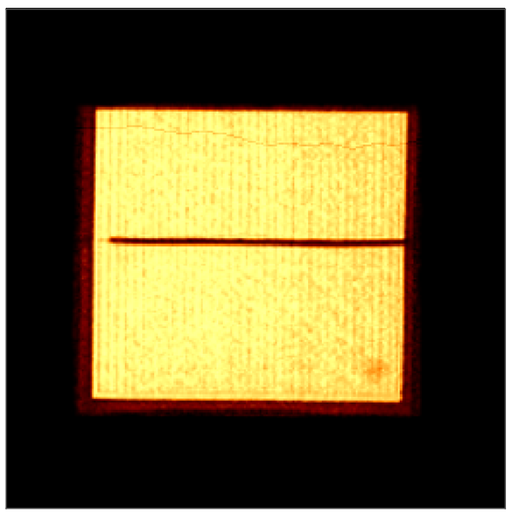
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

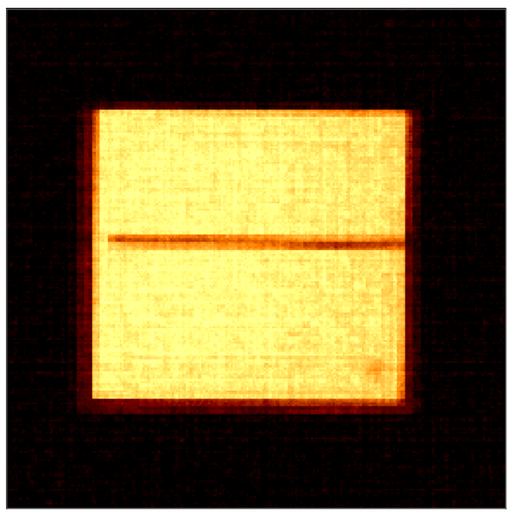
# Bad images for UTV | Reference_0793

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

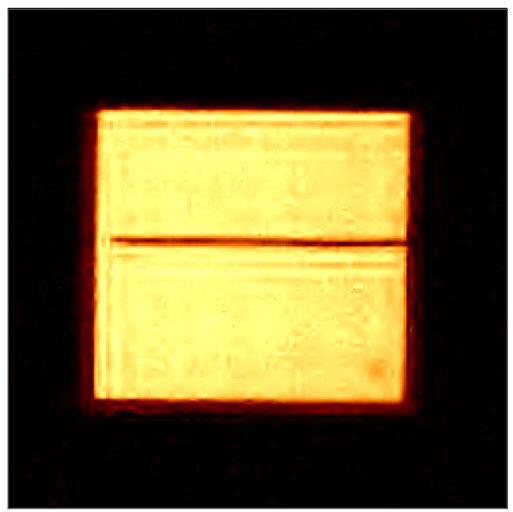
Ground truth
Ground truth



Zero-fill
PSNR 24.50 dB | SSIM 0.416
HaarPSI 0.501



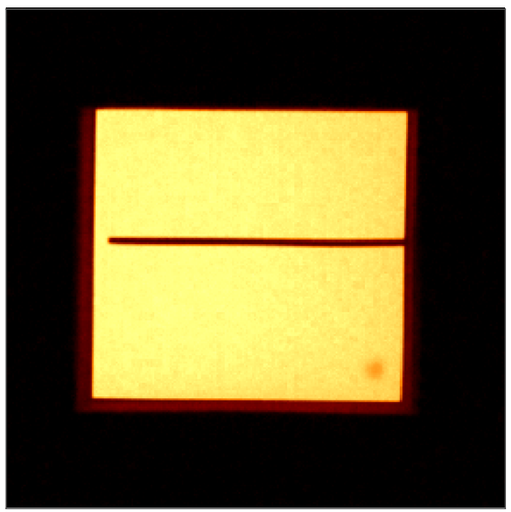
SPGL1-BPDN
PSNR 28.04 dB | SSIM 0.669
HaarPSI 0.580



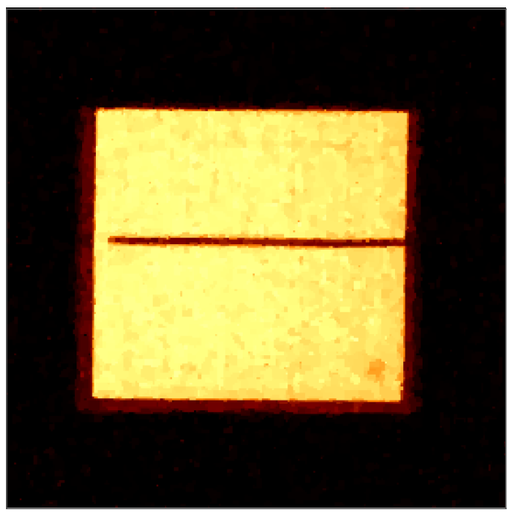
PnP-ADMM-tuned-5
PSNR 33.17 dB | SSIM 0.854
HaarPSI 0.778



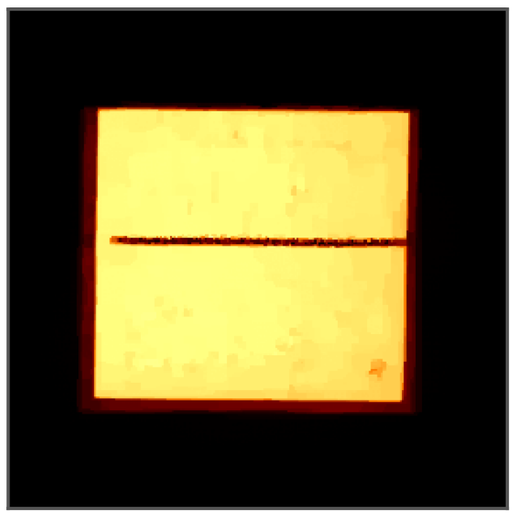
GS-PnP-tuned-5
PSNR 30.74 dB | SSIM 0.679
HaarPSI 0.718



Scalar TV
PSNR 29.78 dB | SSIM 0.634
HaarPSI 0.715



UTV C60
PSNR 31.46 dB | SSIM 0.782
HaarPSI 0.683






## Comparison notes

All methods share the exact frozen measurement and sampling indices.

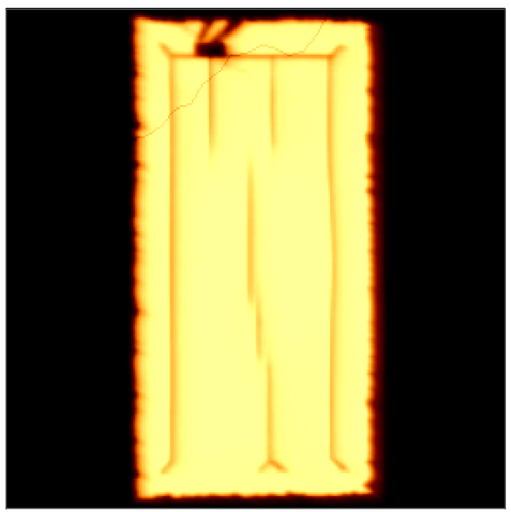
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Visual observation: all compared methods failed.

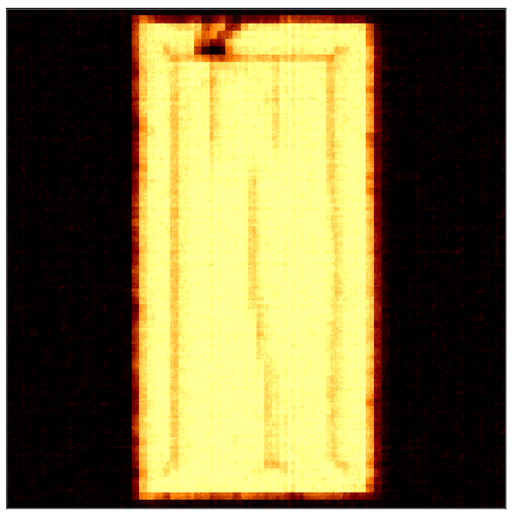
# Bad images for UTV | CIGS_0254

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

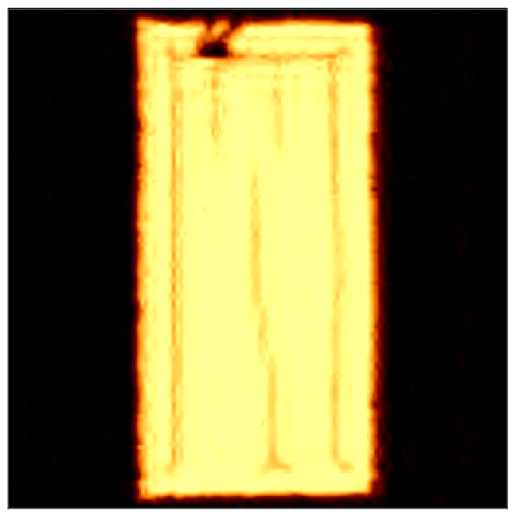
Ground truth
Ground truth



Zero-fill
PSNR 26.23 dB | SSIM 0.507
HaarPSI 0.603



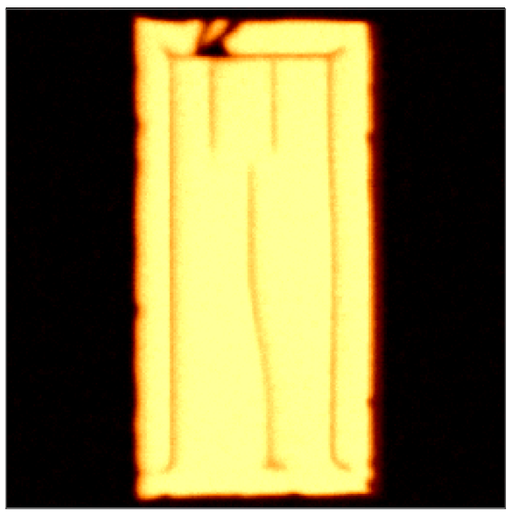
SPGL1-BPDN
PSNR 32.11 dB | SSIM 0.824
HaarPSI 0.764



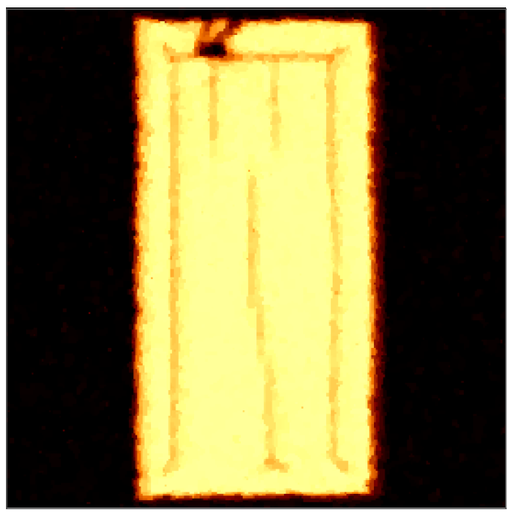
PnP-ADMM-tuned-5
PSNR 39.50 dB | SSIM 0.983
HaarPSI 0.947



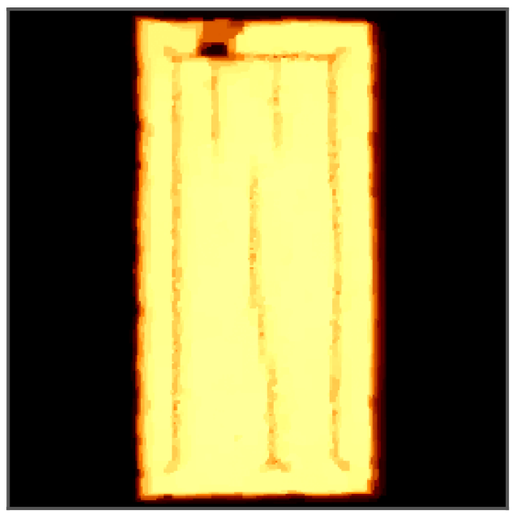
GS-PnP-tuned-5
PSNR 34.12 dB | SSIM 0.806
HaarPSI 0.887



Scalar TV
PSNR 31.72 dB | SSIM 0.766
HaarPSI 0.845



UTV C60
PSNR 33.34 dB | SSIM 0.938
HaarPSI 0.825






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

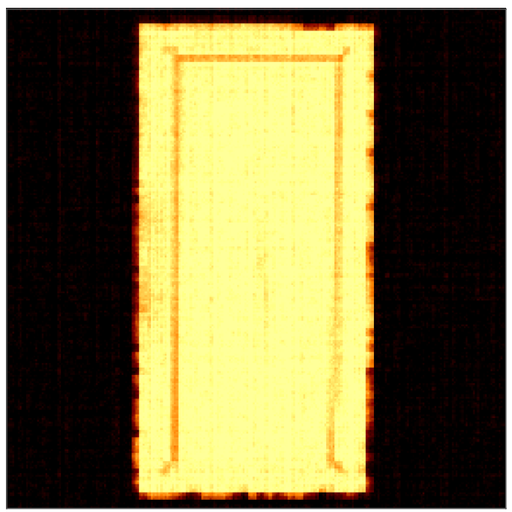
# Bad images for UTV | CIGS_0710

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

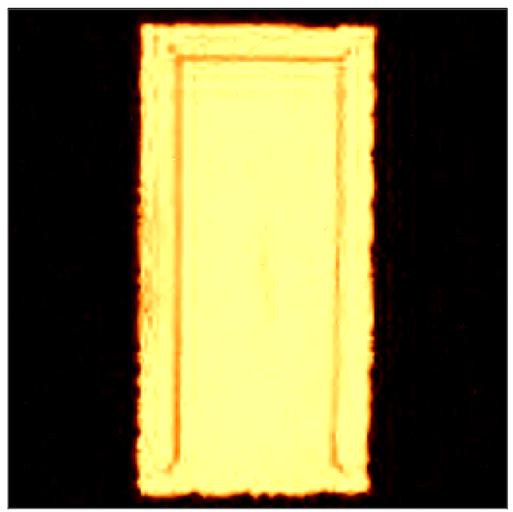
Ground truth
Ground truth



Zero-fill
PSNR 24.80 dB | SSIM 0.411
HaarPSI 0.493



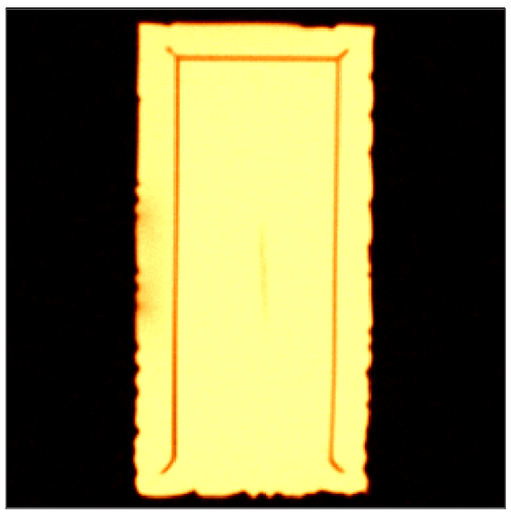
SPGL1-BPDN
PSNR 28.87 dB | SSIM 0.756
HaarPSI 0.587



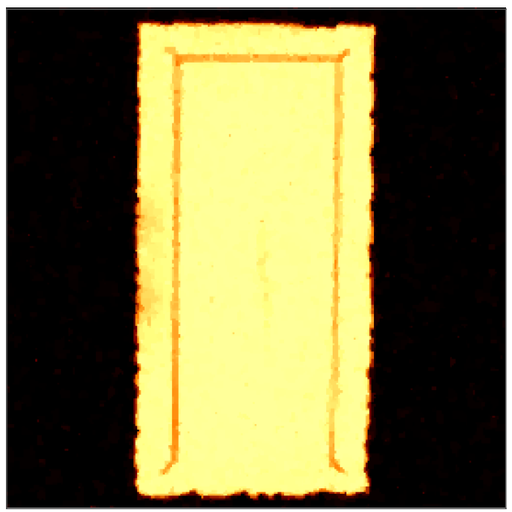
PnP-ADMM-tuned-5
PSNR 37.25 dB | SSIM 0.985
HaarPSI 0.937



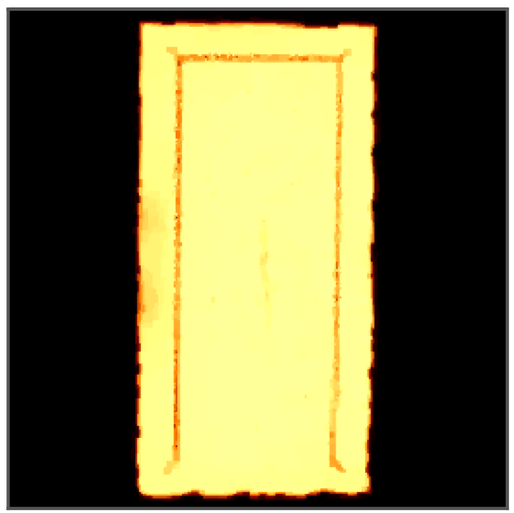
GS-PnP-tuned-5
PSNR 34.66 dB | SSIM 0.864
HaarPSI 0.895



Scalar TV
PSNR 30.48 dB | SSIM 0.738
HaarPSI 0.801



UTV C60
PSNR 32.60 dB | SSIM 0.957
HaarPSI 0.875






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

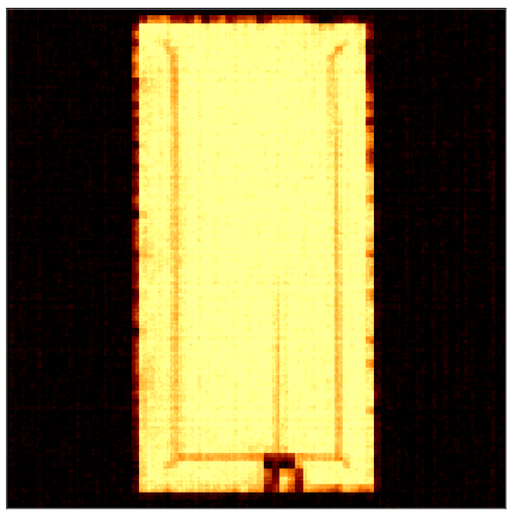
# Bad images for UTV | CIGS_0727

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

Ground truth
Ground truth



Zero-fill
PSNR 24.31 dB | SSIM 0.417
HaarPSI 0.489



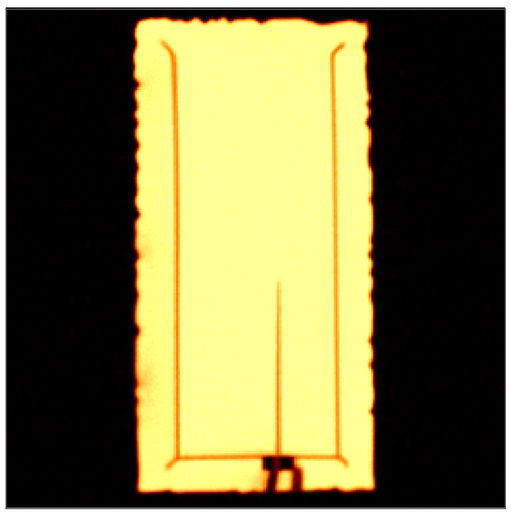
SPGL1-BPDN
PSNR 27.62 dB | SSIM 0.712
HaarPSI 0.555



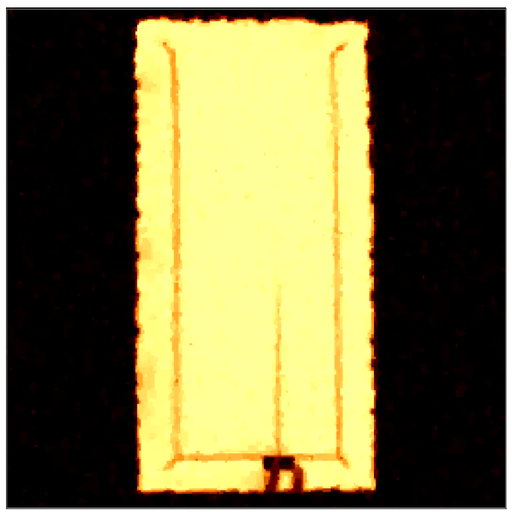
PnP-ADMM-tuned-5
PSNR 36.98 dB | SSIM 0.978
HaarPSI 0.927



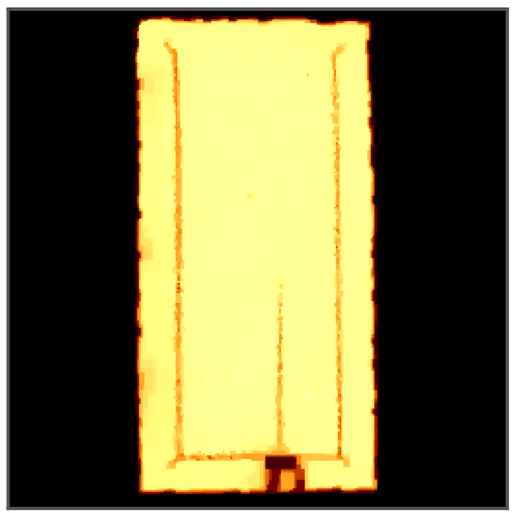
GS-PnP-tuned-5
PSNR 34.46 dB | SSIM 0.865
HaarPSI 0.891



Scalar TV
PSNR 30.12 dB | SSIM 0.726
HaarPSI 0.797



UTV C60
PSNR 31.51 dB | SSIM 0.945
HaarPSI 0.856






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

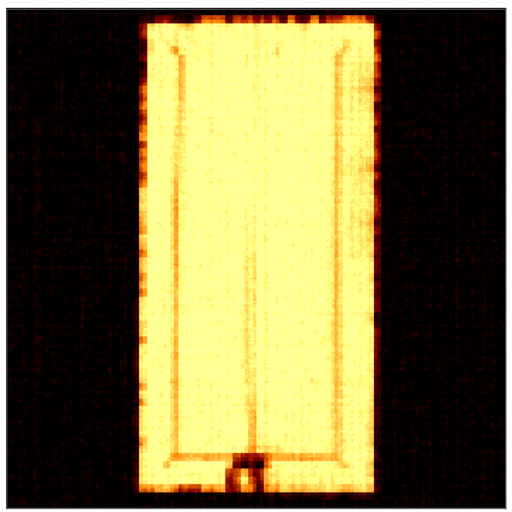
# Bad images for UTV | CIGS_0805

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

Ground truth
Ground truth



Zero-fill
PSNR 24.34 dB | SSIM 0.419
HaarPSI 0.500



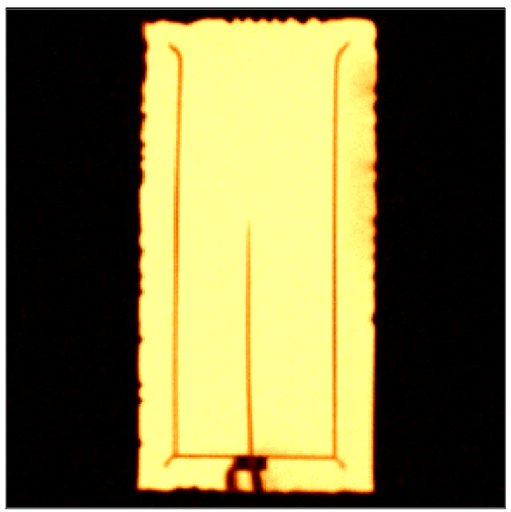
SPGL1-BPDN
PSNR 28.09 dB | SSIM 0.725
HaarPSI 0.554



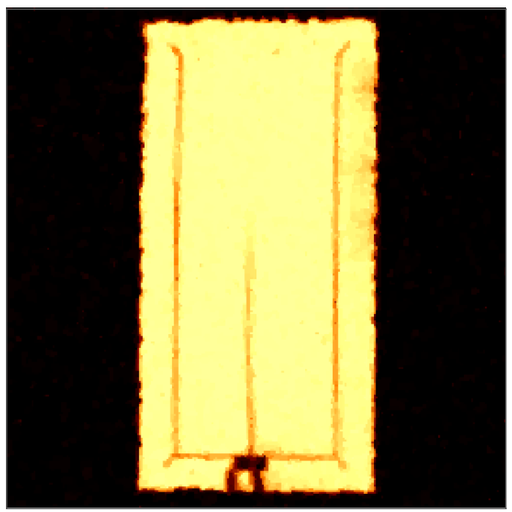
PnP-ADMM-tuned-5
PSNR 36.81 dB | SSIM 0.982
HaarPSI 0.926



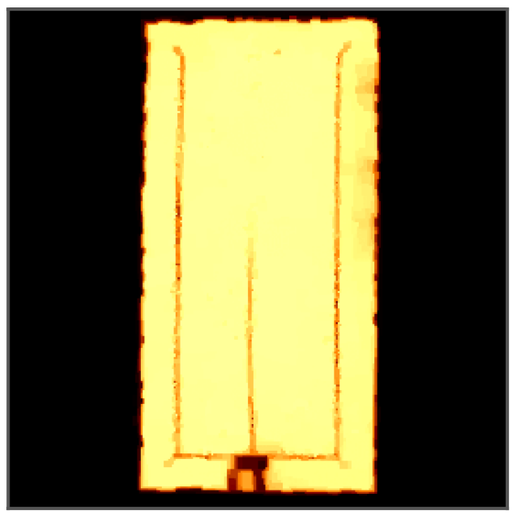
GS-PnP-tuned-5
PSNR 33.19 dB | SSIM 0.790
HaarPSI 0.866



Scalar TV
PSNR 30.41 dB | SSIM 0.745
HaarPSI 0.807



UTV C60
PSNR 31.65 dB | SSIM 0.951
HaarPSI 0.868






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

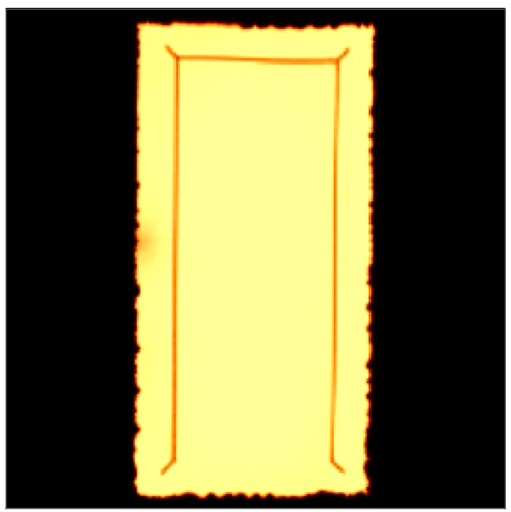
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

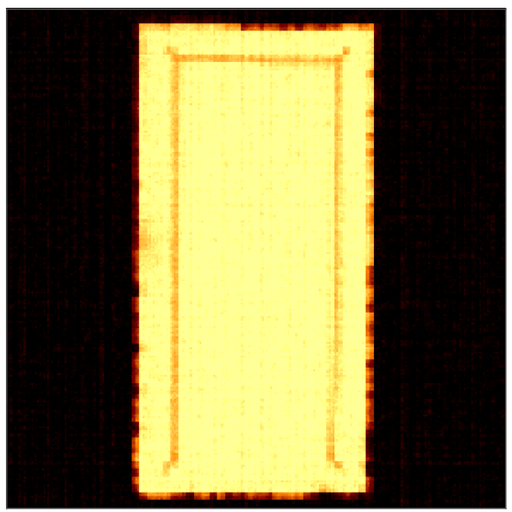
# Bad images for UTV | CIGS_0821

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

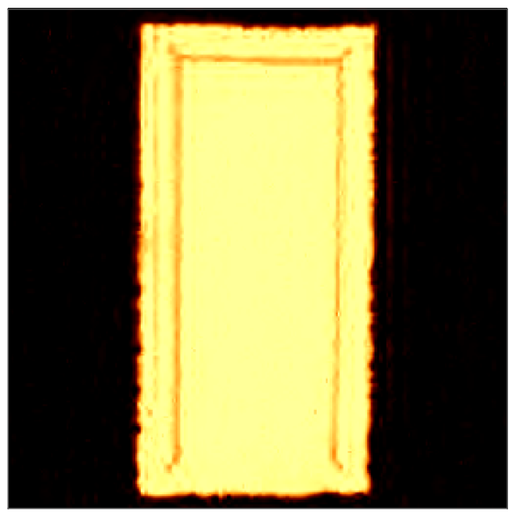
Ground truth
Ground truth



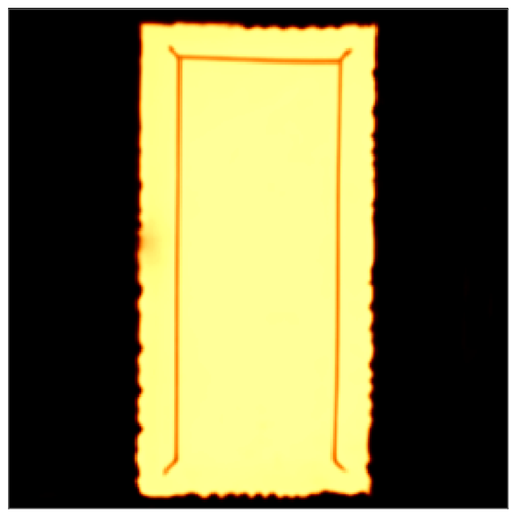
Zero-fill
PSNR 24.52 dB | SSIM 0.403
HaarPSI 0.492



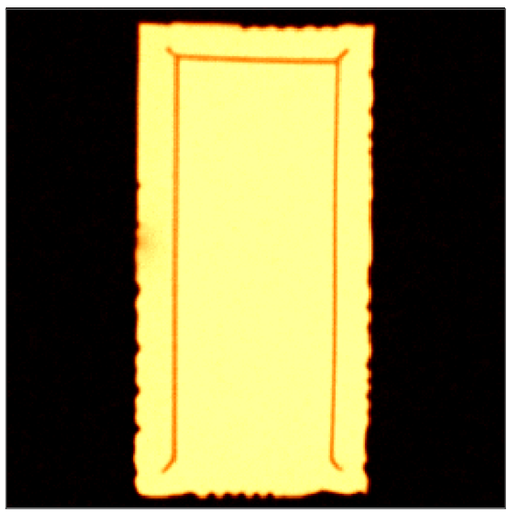
SPGL1-BPDN
PSNR 28.31 dB | SSIM 0.735
HaarPSI 0.536



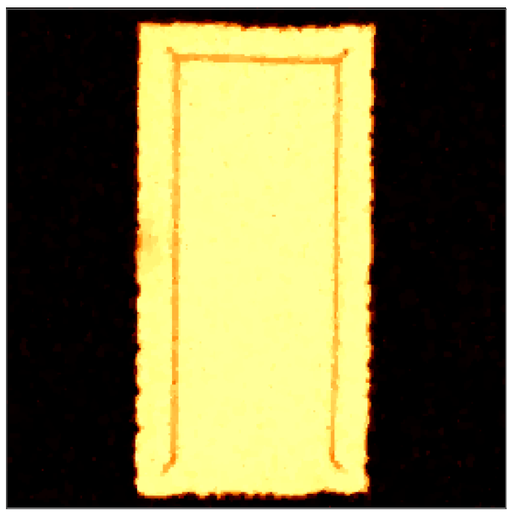
PnP-ADMM-tuned-5
PSNR 37.80 dB | SSIM 0.987
HaarPSI 0.938



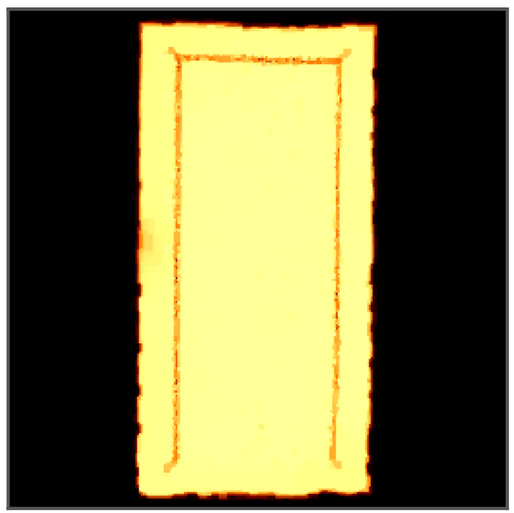
GS-PnP-tuned-5
PSNR 33.60 dB | SSIM 0.791
HaarPSI 0.873



Scalar TV
PSNR 30.95 dB | SSIM 0.741
HaarPSI 0.808



UTV C60
PSNR 32.53 dB | SSIM 0.960
HaarPSI 0.892






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

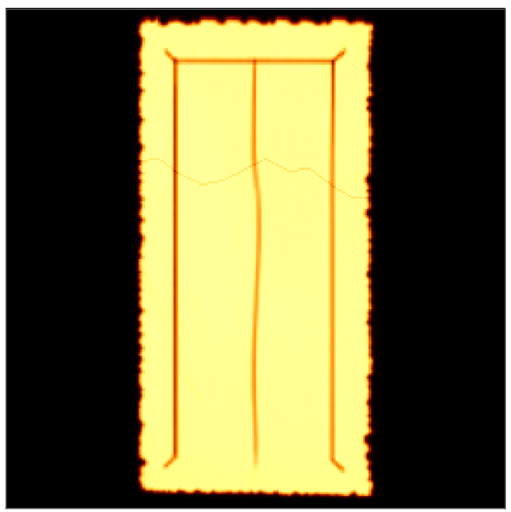
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

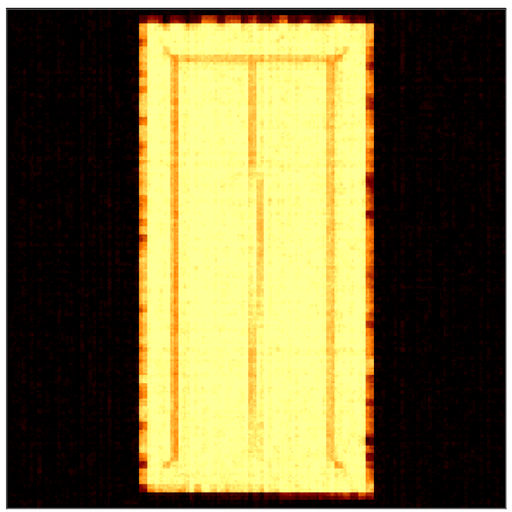
# Bad images for UTV | CIGS_0873

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

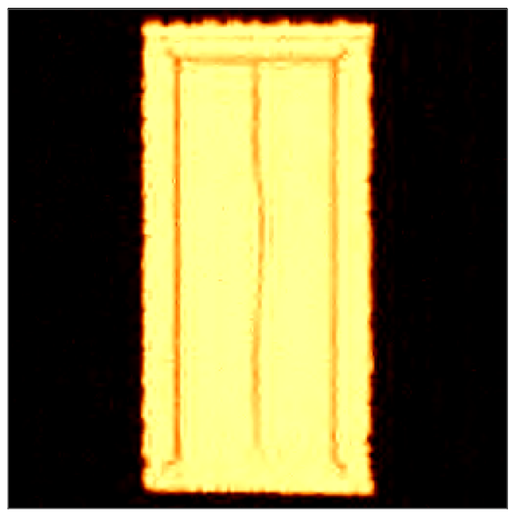
Ground truth
Ground truth



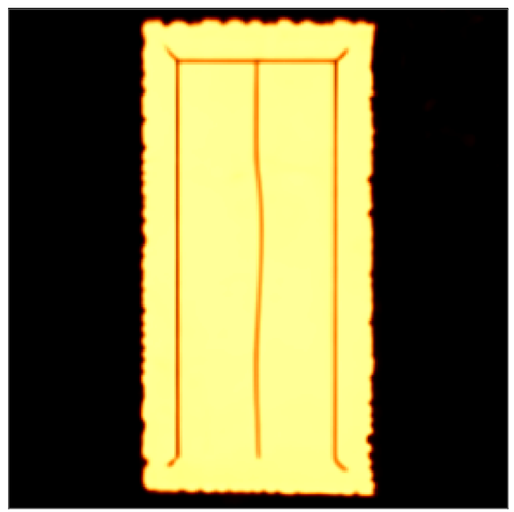
Zero-fill
PSNR 23.61 dB | SSIM 0.416
HaarPSI 0.458



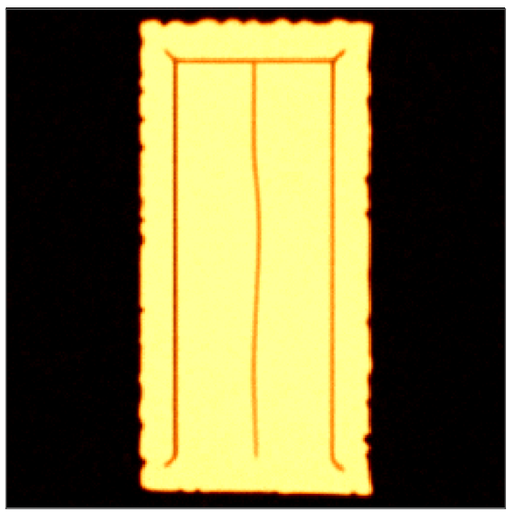
SPGL1-BPDN
PSNR 28.43 dB | SSIM 0.740
HaarPSI 0.591



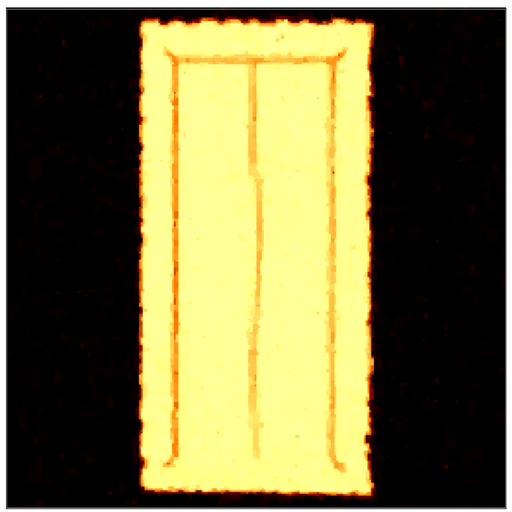
PnP-ADMM-tuned-5
PSNR 37.82 dB | SSIM 0.979
HaarPSI 0.946



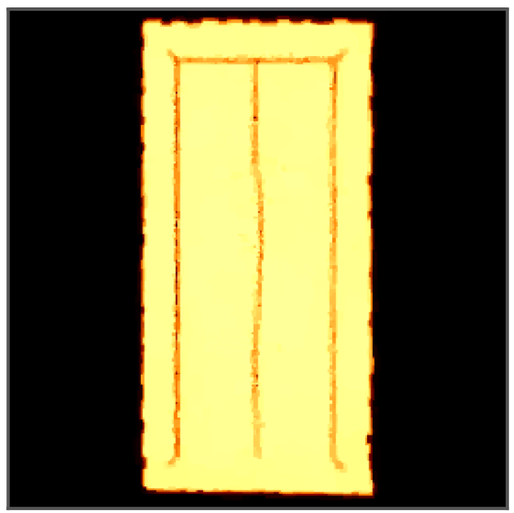
GS-PnP-tuned-5
PSNR 33.39 dB | SSIM 0.789
HaarPSI 0.874



Scalar TV
PSNR 29.47 dB | SSIM 0.727
HaarPSI 0.767



UTV C60
PSNR 31.32 dB | SSIM 0.948
HaarPSI 0.857






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

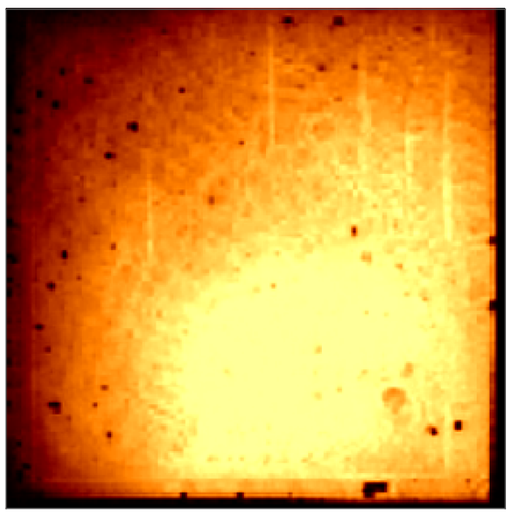
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

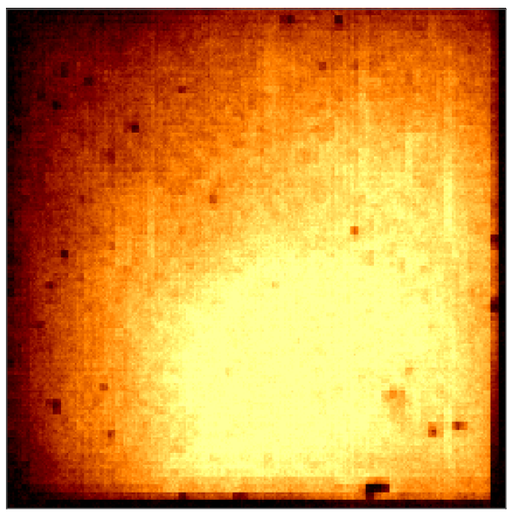
# Bad images for UTV | Perovskite_0021

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

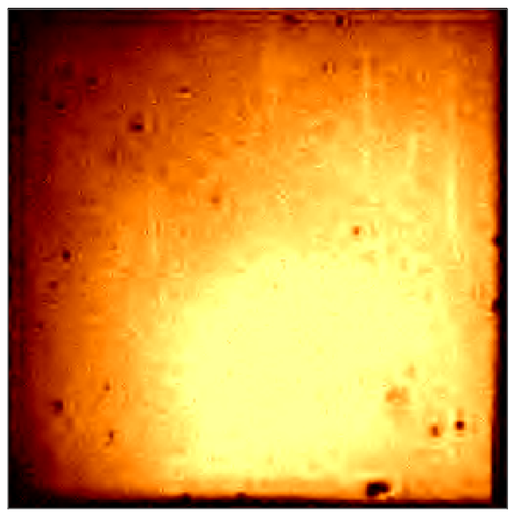
Ground truth
Ground truth



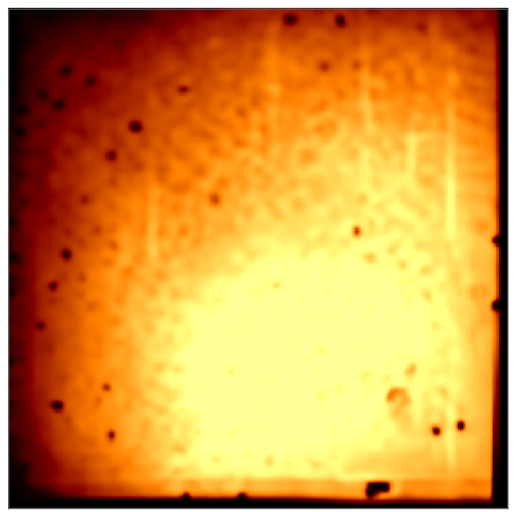
Zero-fill
PSNR 27.54 dB | SSIM 0.619
HaarPSI 0.626



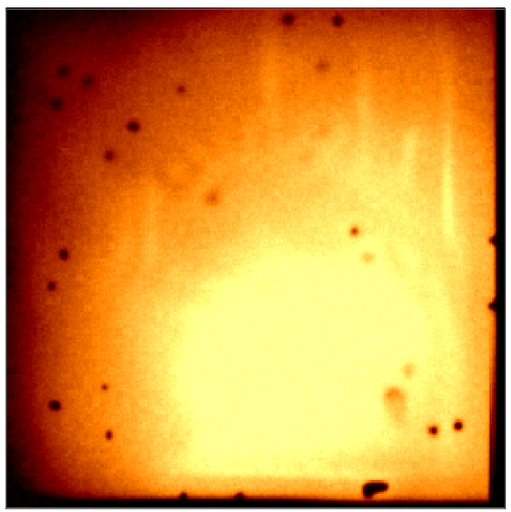
SPGL1-BPDN
PSNR 30.01 dB | SSIM 0.767
HaarPSI 0.662



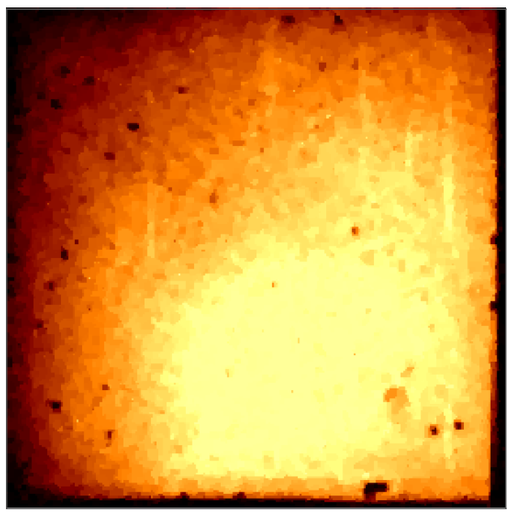
PnP-ADMM-tuned-5
PSNR 33.50 dB | SSIM 0.866
HaarPSI 0.801



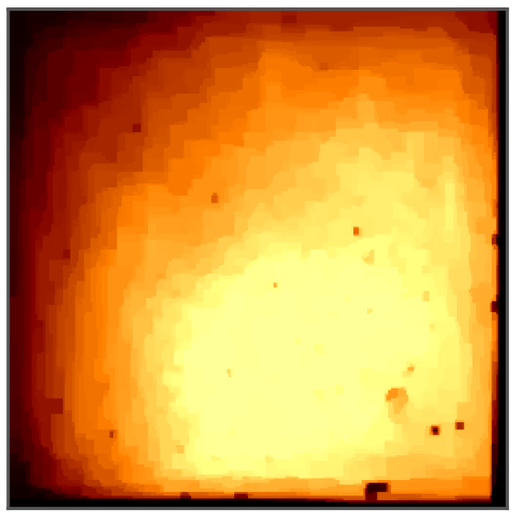
GS-PnP-tuned-5
PSNR 29.97 dB | SSIM 0.668
HaarPSI 0.676



Scalar TV
PSNR 31.04 dB | SSIM 0.785
HaarPSI 0.744



UTV C60
PSNR 30.95 dB | SSIM 0.796
HaarPSI 0.623






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

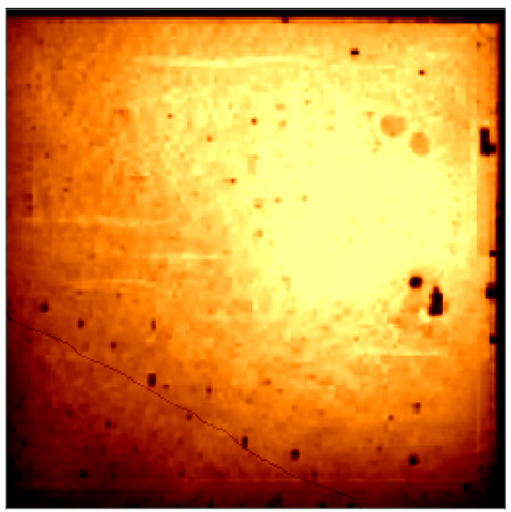
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

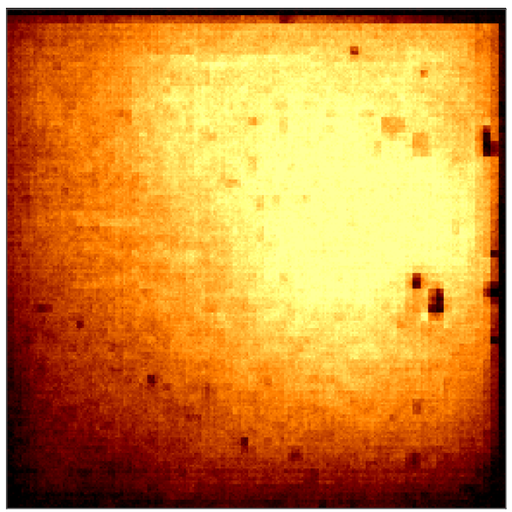
# Bad images for UTV | Perovskite_0031

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

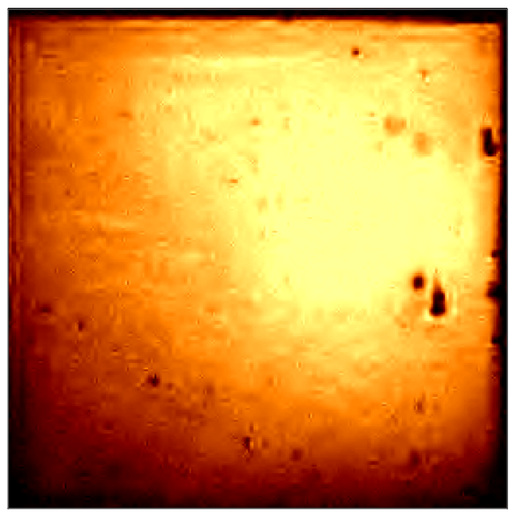
Ground truth
Ground truth



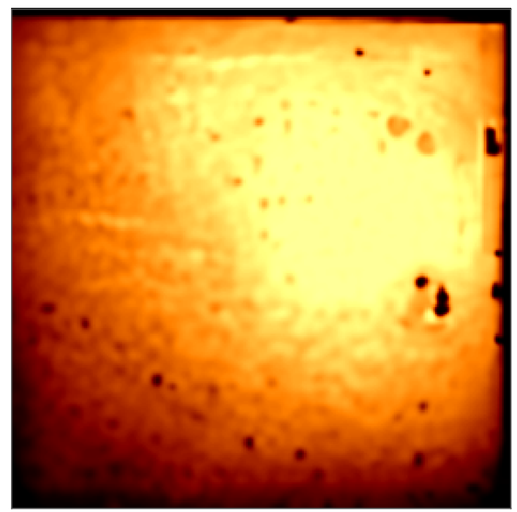
Zero-fill
PSNR 27.19 dB | SSIM 0.606
HaarPSI 0.616



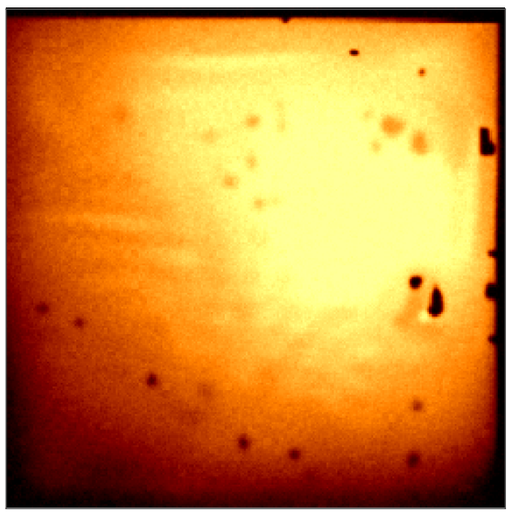
SPGL1-BPDN
PSNR 29.82 dB | SSIM 0.751
HaarPSI 0.685



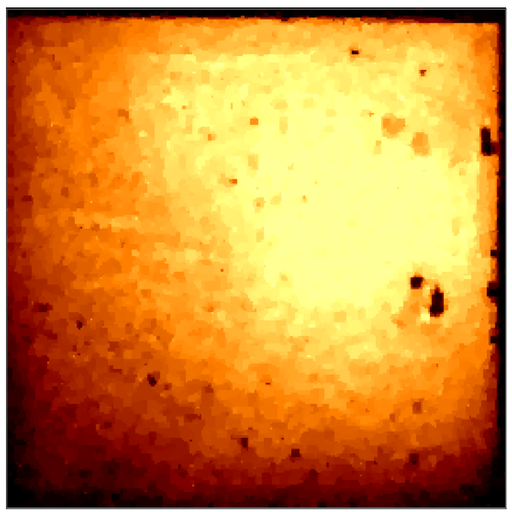
PnP-ADMM-tuned-5
PSNR 33.00 dB | SSIM 0.849
HaarPSI 0.789



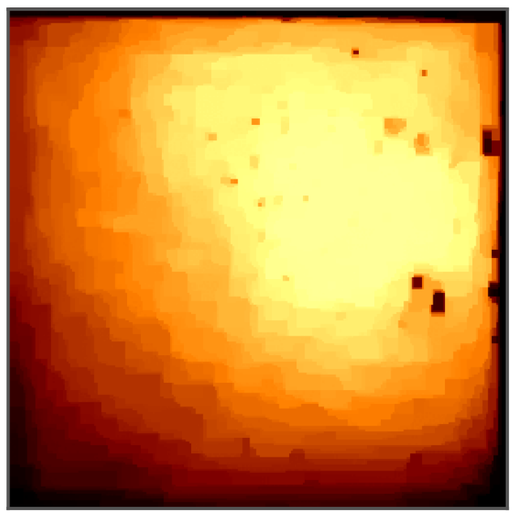
GS-PnP-tuned-5
PSNR 29.68 dB | SSIM 0.652
HaarPSI 0.662



Scalar TV
PSNR 30.68 dB | SSIM 0.773
HaarPSI 0.745



UTV C60
PSNR 30.46 dB | SSIM 0.780
HaarPSI 0.615






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

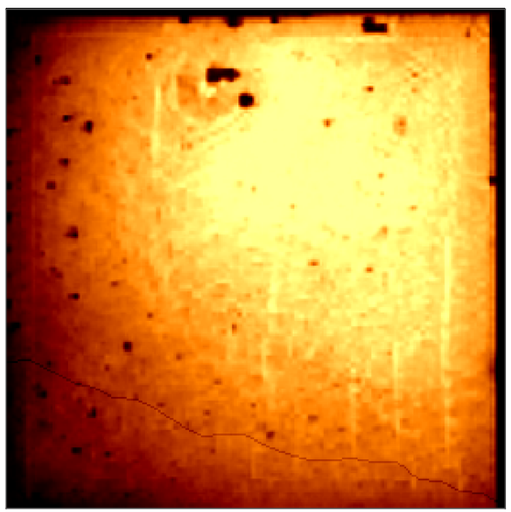
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

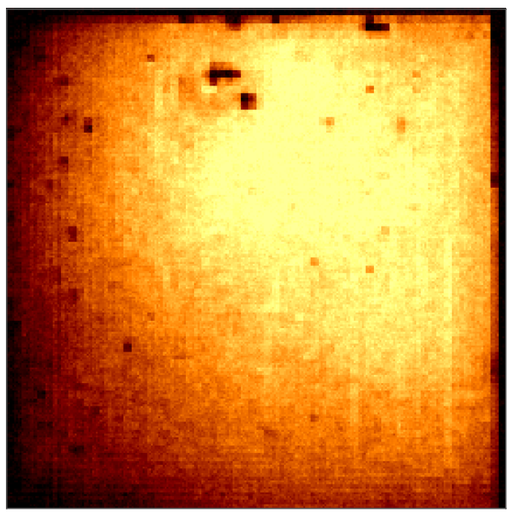
# Bad images for UTV | Perovskite_0079

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

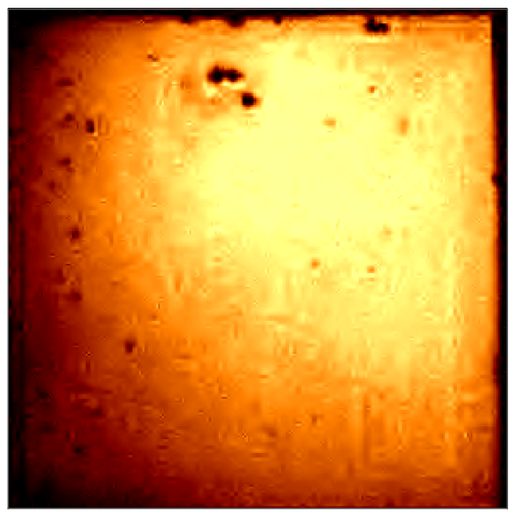
Ground truth
Ground truth



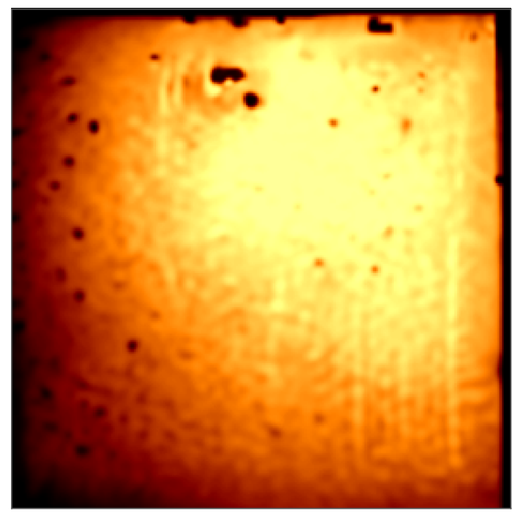
Zero-fill
PSNR 27.03 dB | SSIM 0.608
HaarPSI 0.618



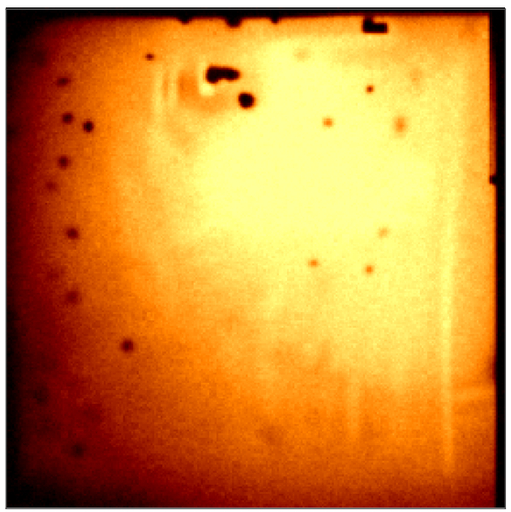
SPGL1-BPDN
PSNR 29.42 dB | SSIM 0.746
HaarPSI 0.661



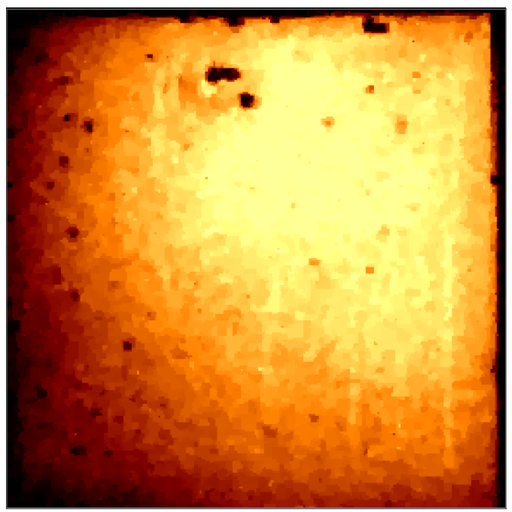
PnP-ADMM-tuned-5
PSNR 32.73 dB | SSIM 0.846
HaarPSI 0.787



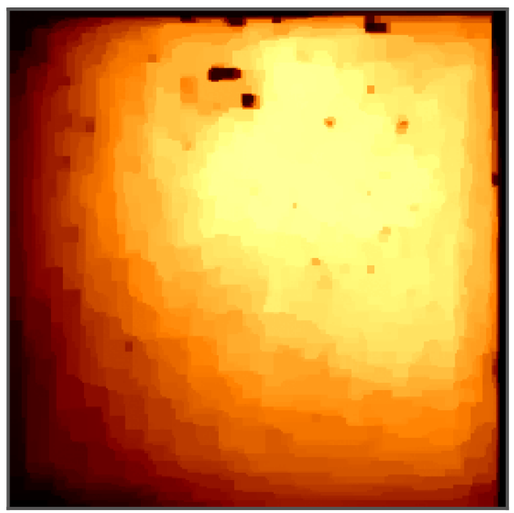
GS-PnP-tuned-5
PSNR 29.45 dB | SSIM 0.640
HaarPSI 0.661



Scalar TV
PSNR 30.44 dB | SSIM 0.771
HaarPSI 0.729



UTV C60
PSNR 29.98 dB | SSIM 0.762
HaarPSI 0.613






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

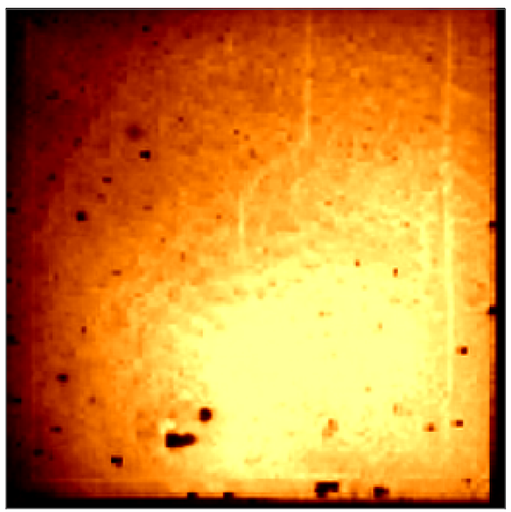
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

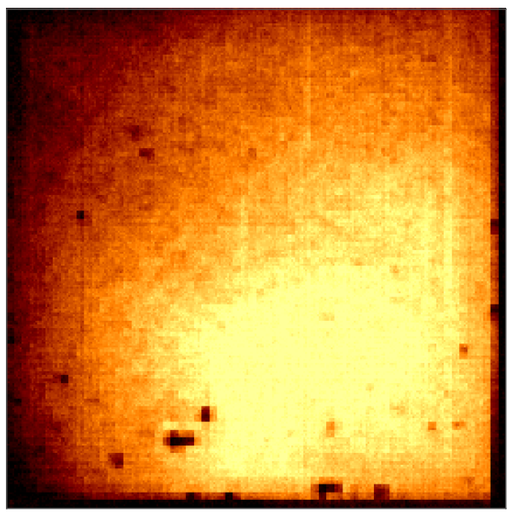
# Bad images for UTV | Perovskite_0098

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

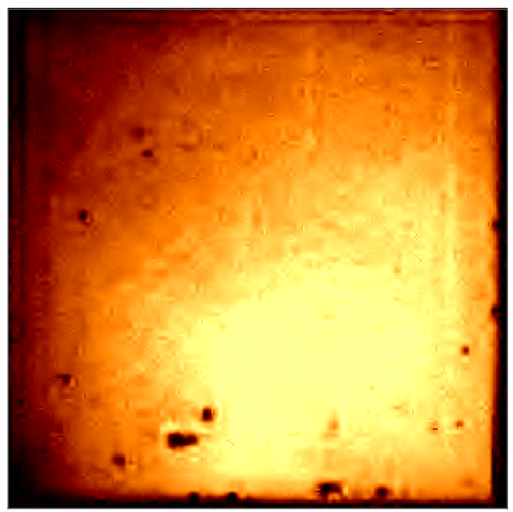
Ground truth
Ground truth



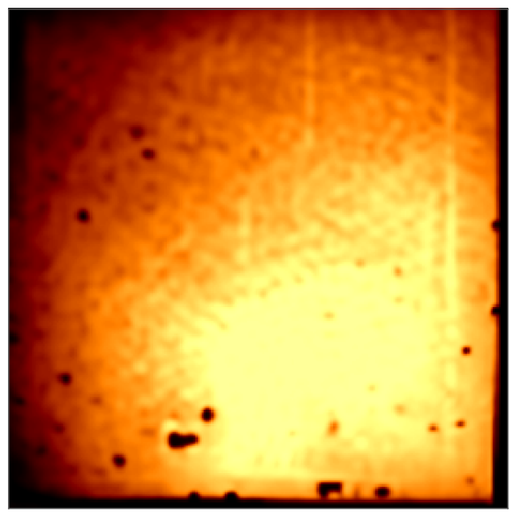
Zero-fill
PSNR 27.22 dB | SSIM 0.608
HaarPSI 0.624



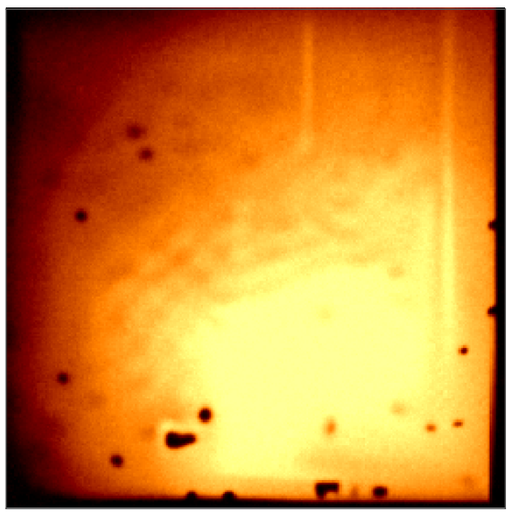
SPGL1-BPDN
PSNR 29.65 dB | SSIM 0.755
HaarPSI 0.646



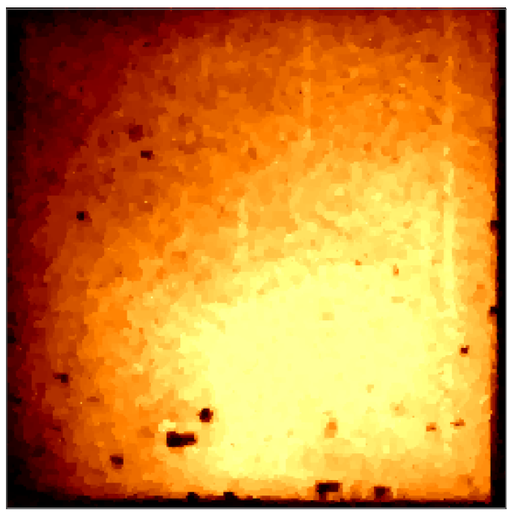
PnP-ADMM-tuned-5
PSNR 32.91 dB | SSIM 0.854
HaarPSI 0.785



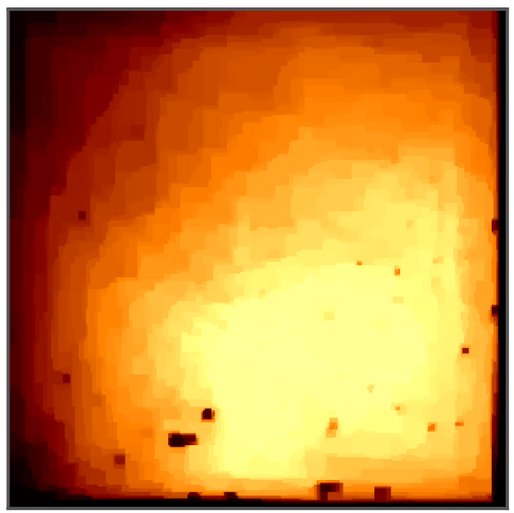
GS-PnP-tuned-5
PSNR 30.52 dB | SSIM 0.715
HaarPSI 0.684



Scalar TV
PSNR 30.72 dB | SSIM 0.781
HaarPSI 0.729



UTV C60
PSNR 30.46 dB | SSIM 0.780
HaarPSI 0.616






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

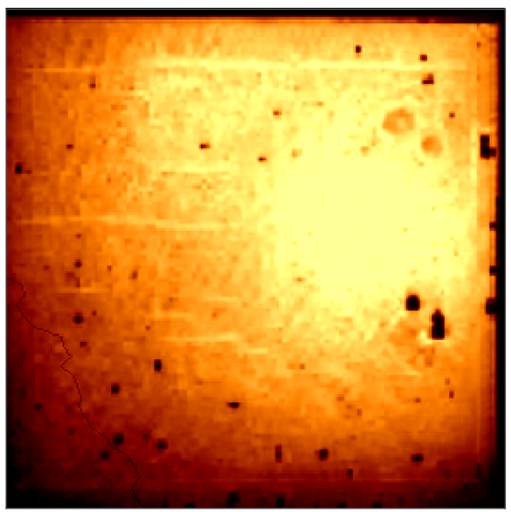
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

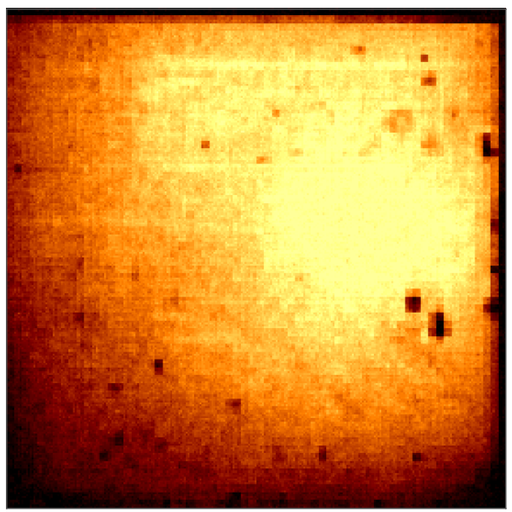
# Bad images for UTV | Perovskite_0187

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

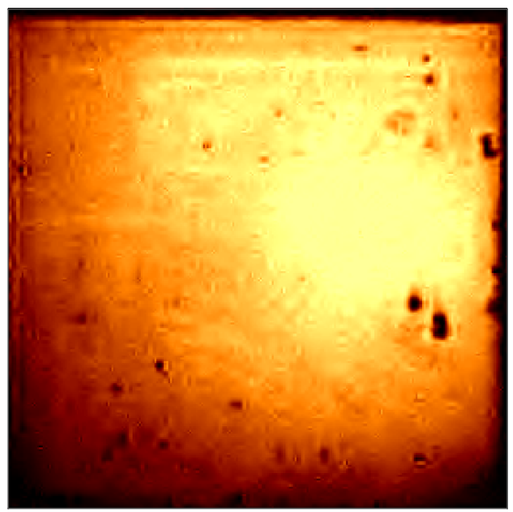
Ground truth
Ground truth



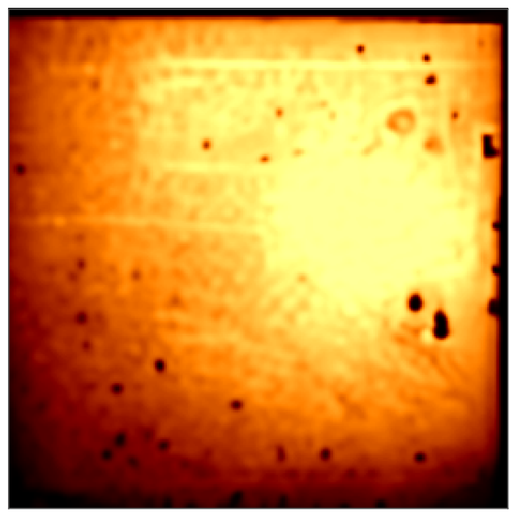
Zero-fill
PSNR 26.88 dB | SSIM 0.615
HaarPSI 0.612



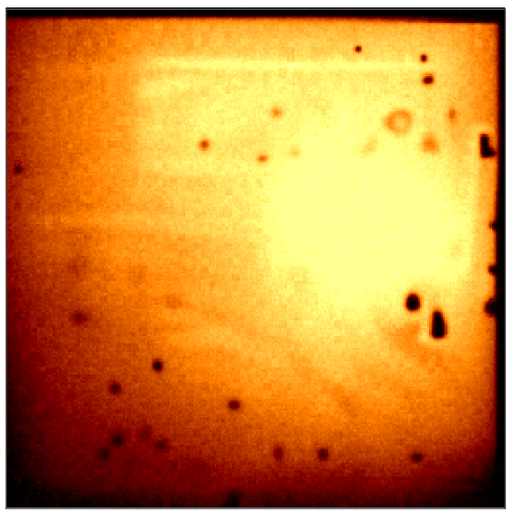
SPGL1-BPDN
PSNR 29.25 dB | SSIM 0.745
HaarPSI 0.666



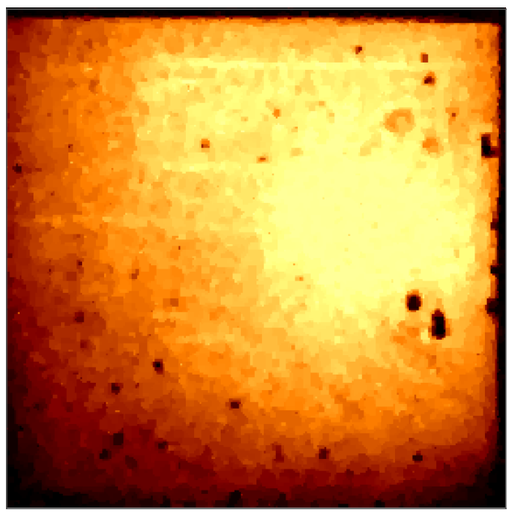
PnP-ADMM-tuned-5
PSNR 32.65 dB | SSIM 0.850
HaarPSI 0.786



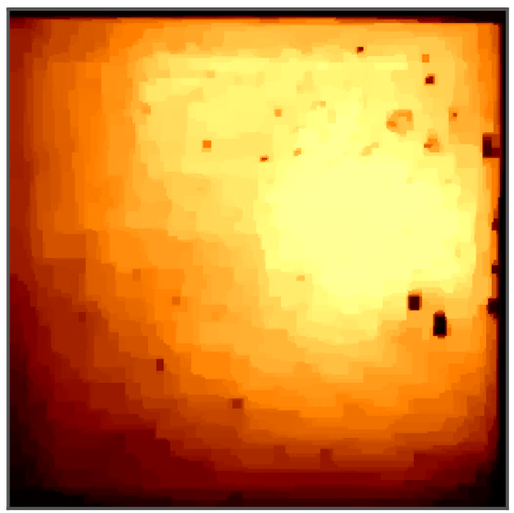
GS-PnP-tuned-5
PSNR 28.50 dB | SSIM 0.584
HaarPSI 0.636



Scalar TV
PSNR 30.43 dB | SSIM 0.776
HaarPSI 0.743



UTV C60
PSNR 29.82 dB | SSIM 0.766
HaarPSI 0.603






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

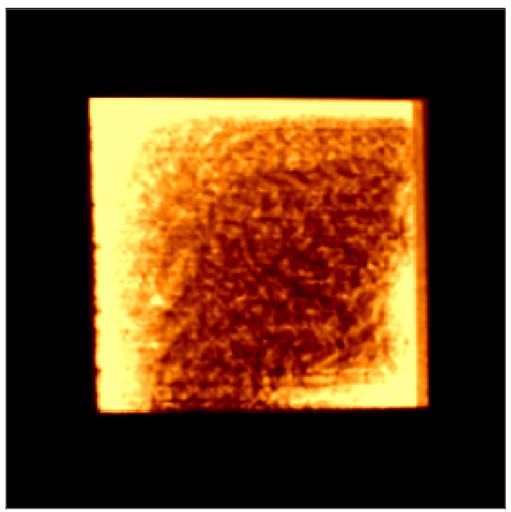
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

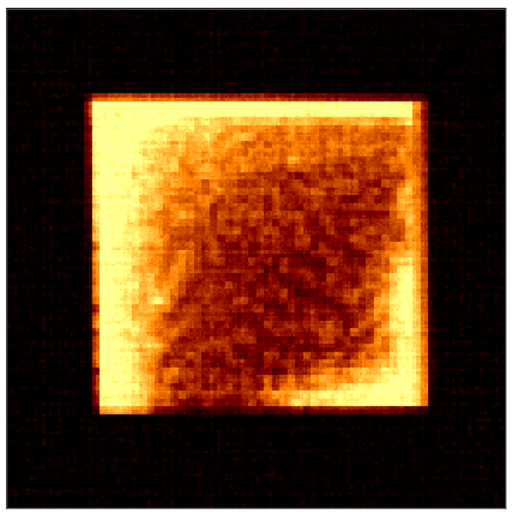
# Bad images for UTV | Perovskite_0260

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

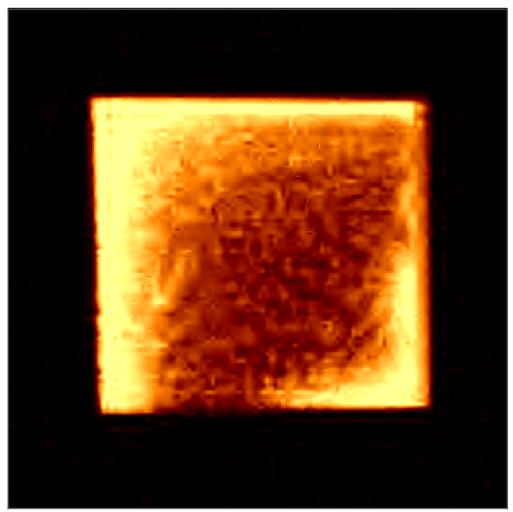
Ground truth
Ground truth



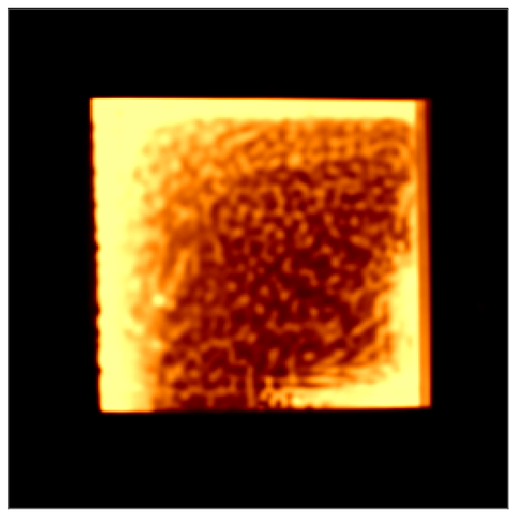
Zero-fill
PSNR 24.51 dB | SSIM 0.473
HaarPSI 0.532



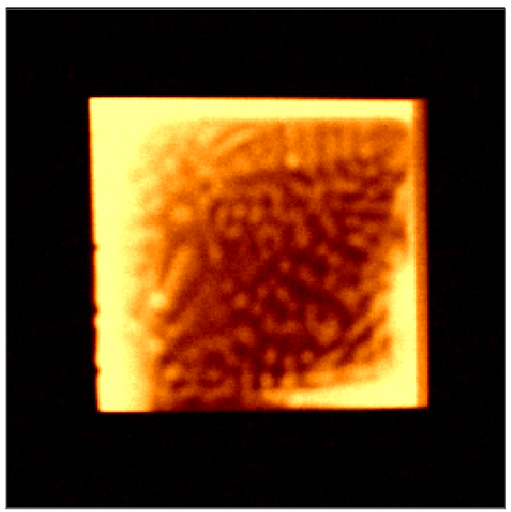
SPGL1-BPDN
PSNR 27.66 dB | SSIM 0.709
HaarPSI 0.580



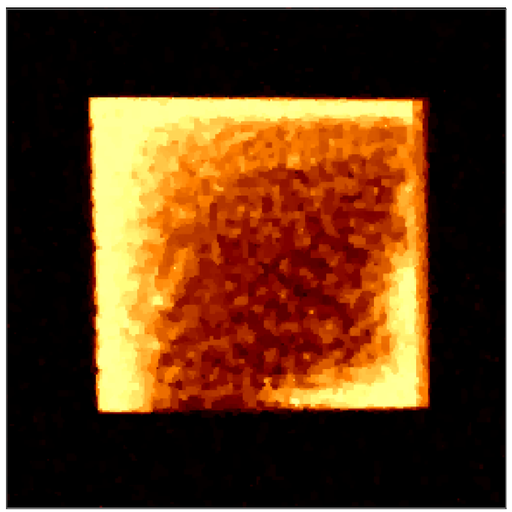
PnP-ADMM-tuned-5
PSNR 32.27 dB | SSIM 0.928
HaarPSI 0.815



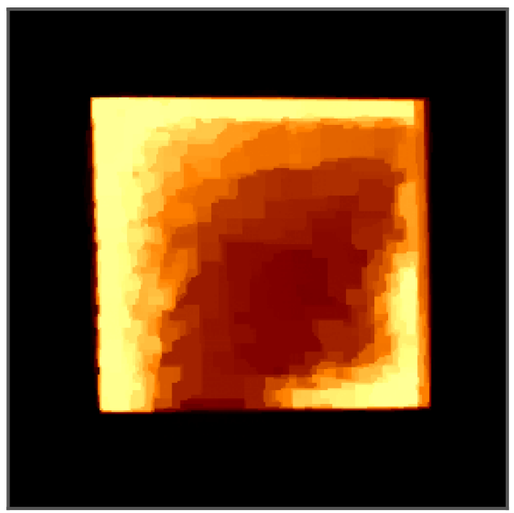
GS-PnP-tuned-5
PSNR 29.34 dB | SSIM 0.711
HaarPSI 0.728



Scalar TV
PSNR 29.79 dB | SSIM 0.731
HaarPSI 0.748



UTV C60
PSNR 28.53 dB | SSIM 0.806
HaarPSI 0.554






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

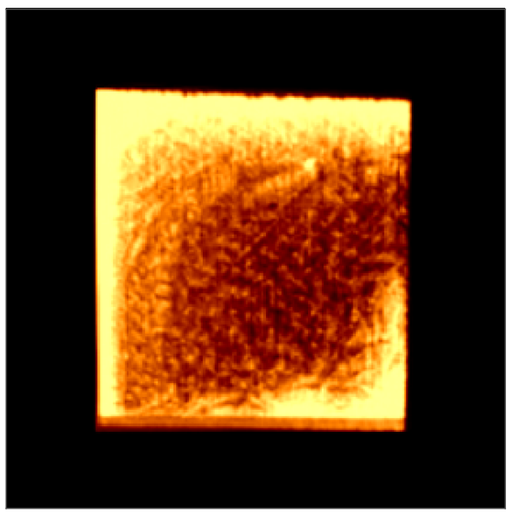
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

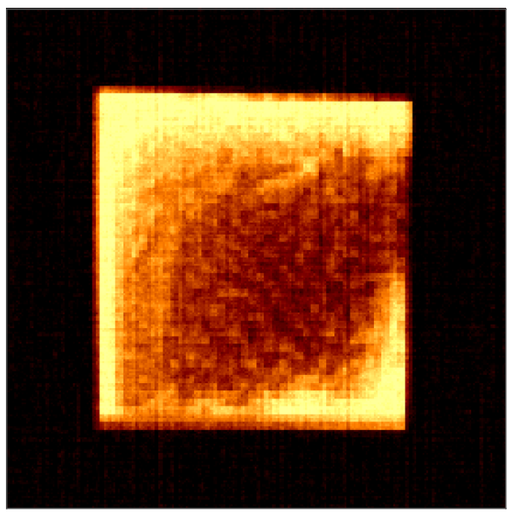
# Bad images for UTV | Perovskite_0288

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

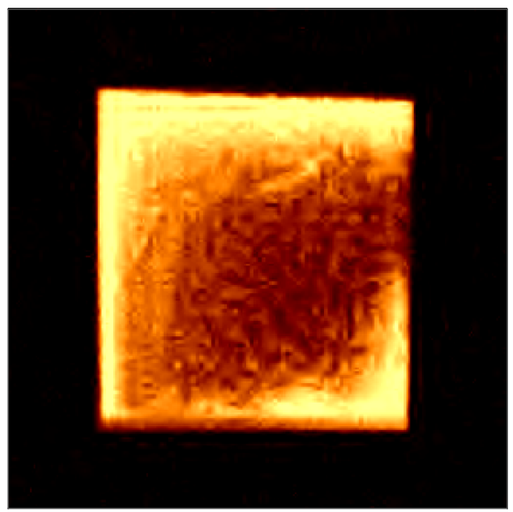
Ground truth
Ground truth



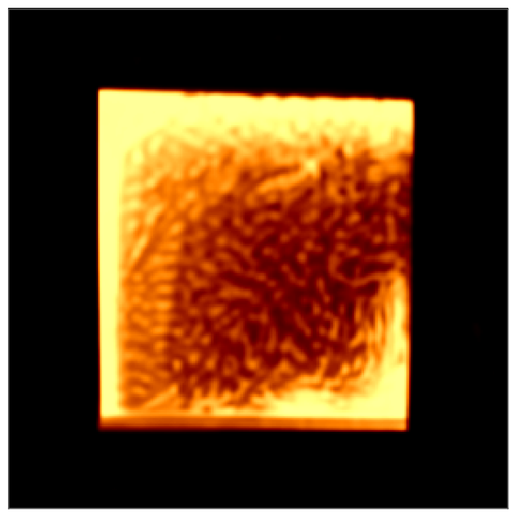
Zero-fill
PSNR 24.69 dB | SSIM 0.464
HaarPSI 0.544



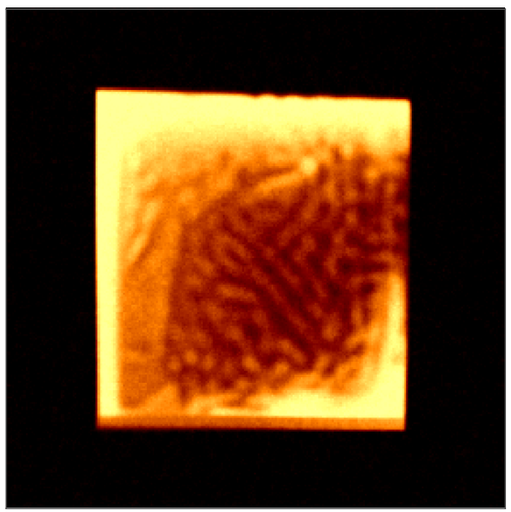
SPGL1-BPDN
PSNR 28.27 dB | SSIM 0.706
HaarPSI 0.605



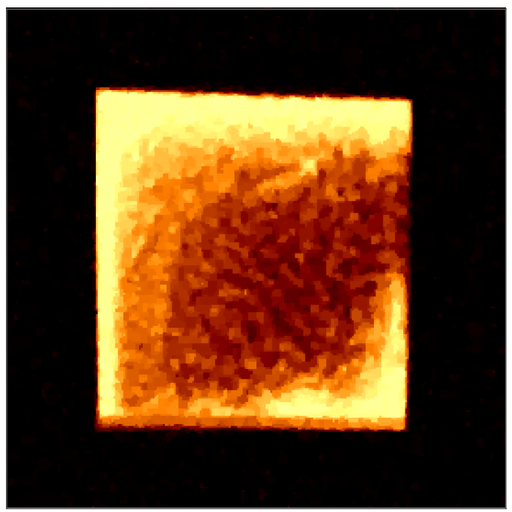
PnP-ADMM-tuned-5
PSNR 32.44 dB | SSIM 0.927
HaarPSI 0.818



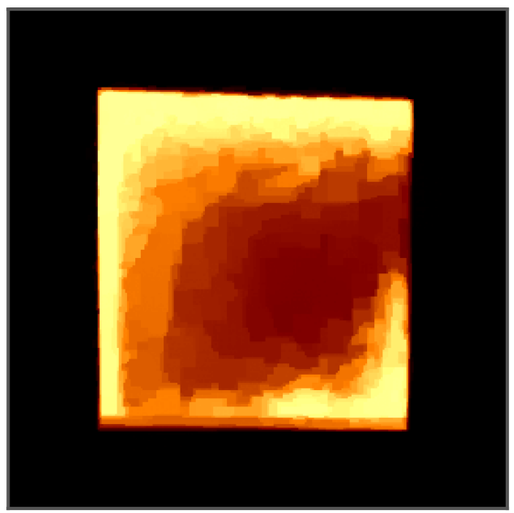
GS-PnP-tuned-5
PSNR 29.27 dB | SSIM 0.698
HaarPSI 0.725



Scalar TV
PSNR 29.88 dB | SSIM 0.726
HaarPSI 0.744



UTV C60
PSNR 28.70 dB | SSIM 0.803
HaarPSI 0.581






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

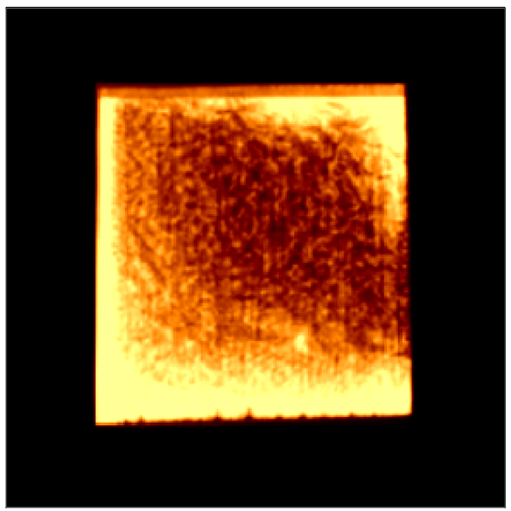
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

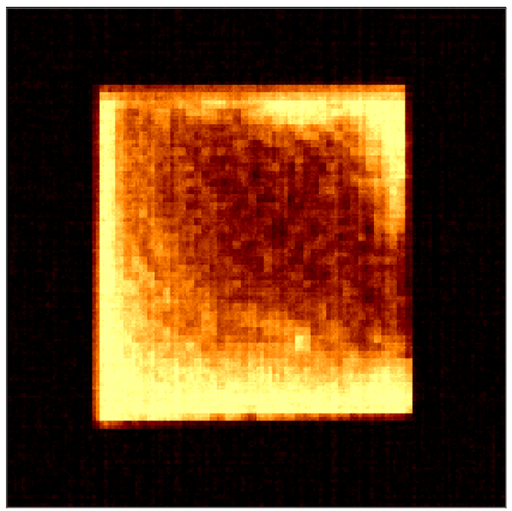
# Bad images for UTV | Perovskite_0292

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

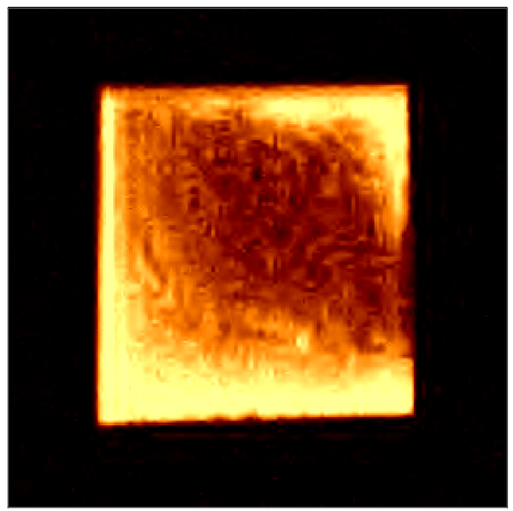
Ground truth
Ground truth



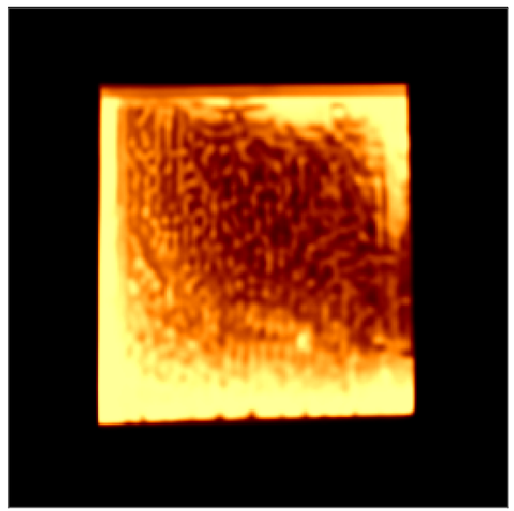
Zero-fill
PSNR 24.57 dB | SSIM 0.476
HaarPSI 0.528



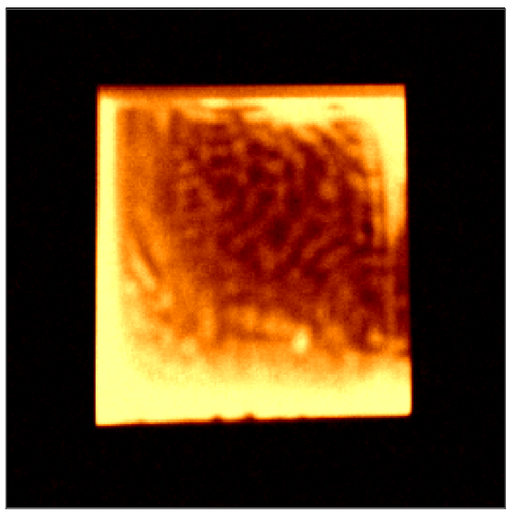
SPGL1-BPDN
PSNR 27.71 dB | SSIM 0.702
HaarPSI 0.591



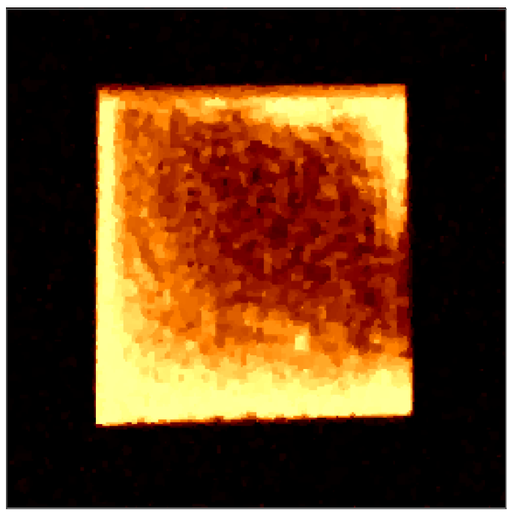
PnP-ADMM-tuned-5
PSNR 32.37 dB | SSIM 0.927
HaarPSI 0.818



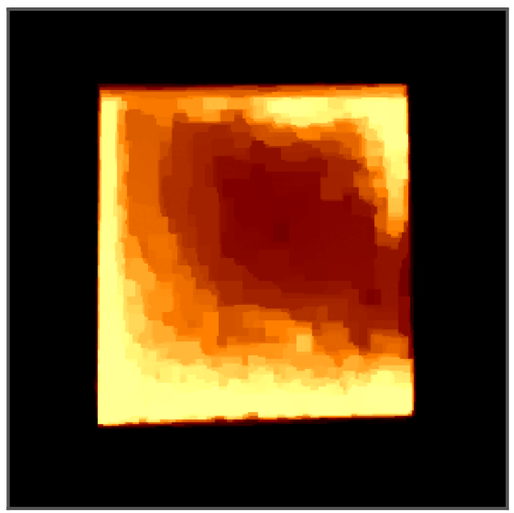
GS-PnP-tuned-5
PSNR 29.20 dB | SSIM 0.688
HaarPSI 0.725



Scalar TV
PSNR 29.79 dB | SSIM 0.717
HaarPSI 0.757



UTV C60
PSNR 28.81 dB | SSIM 0.809
HaarPSI 0.580






Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

All methods share the exact frozen measurement and sampling indices.

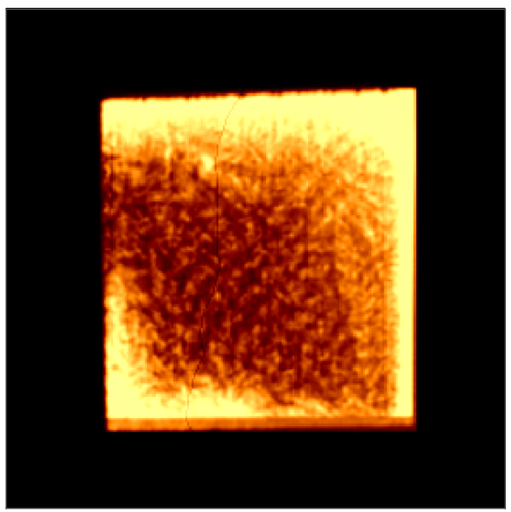
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

# Bad images for UTV | Perovskite_0391

No stored acquisition: ground truth only

Ground truth
Ground truth



Zero-fill
No reconstruction

Not available
(no stored acquisition)

SPGL1-BPDN
No reconstruction

Not available
(no stored acquisition)

PnP-ADMM-tuned-5
No reconstruction

Not available
(no stored acquisition)

GS-PnP-tuned-5
No reconstruction

Not available
(no stored acquisition)

Scalar TV
No reconstruction

Not available
(no stored acquisition)

UTV C60
No reconstruction

Not available
(no stored acquisition)

0.0 0.2 0.4 0.6 0.8 1.0

Common display scale [0, 1]; metrics use unmodified images

## Comparison notes

Perovskite_0391 was not among the frozen 450 test images. Its 20% NPL-default acquisition and the six reconstructions are unavailable.

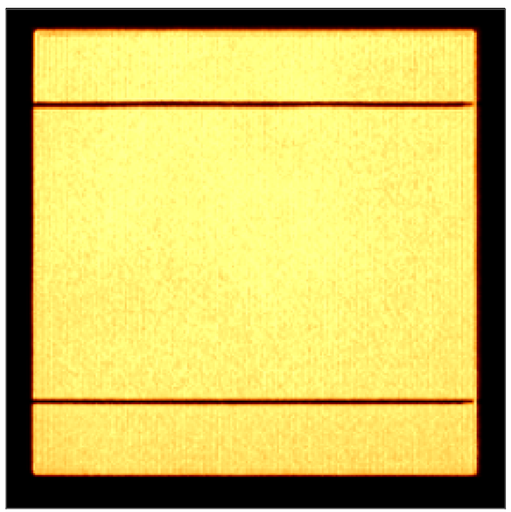
No new noise was simulated.
No metric has been invented.

No acquired measurement

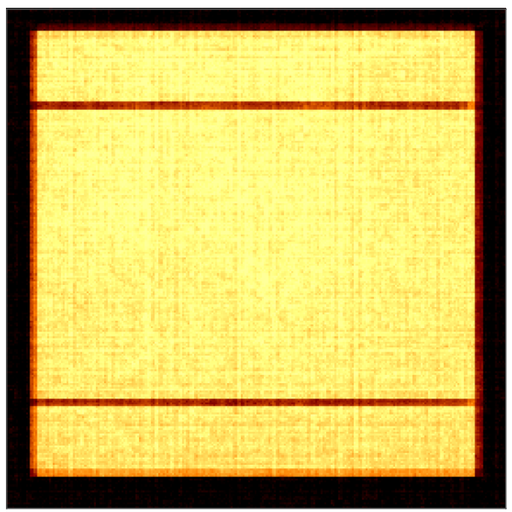
# Bad images for UTV | Silicone_0767

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

Ground truth
Ground truth



Zero-fill
PSNR 22.40 dB | SSIM 0.450
HaarPSI 0.490



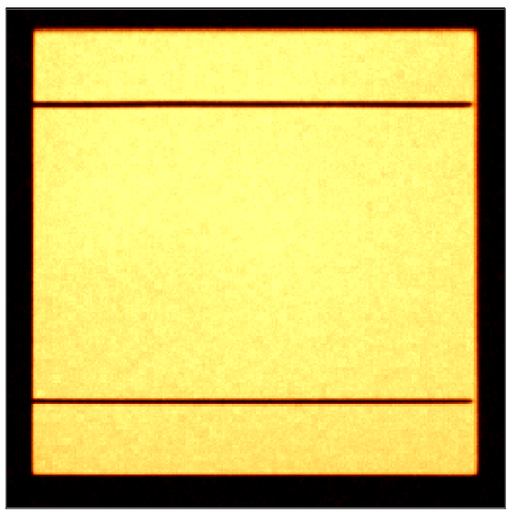
SPGL1-BPDN
PSNR 24.36 dB | SSIM 0.577
HaarPSI 0.503



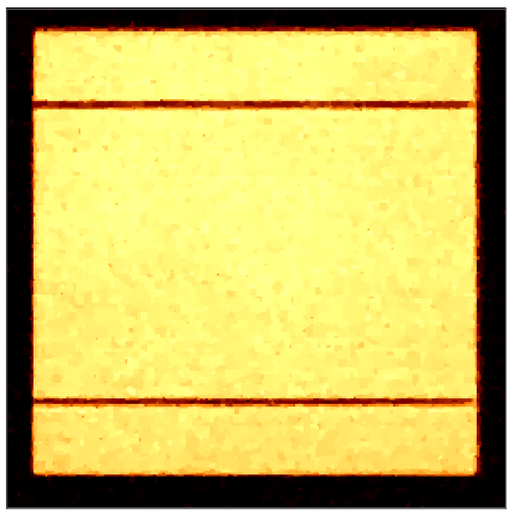
PnP-ADMM-tuned-5
PSNR 33.59 dB | SSIM 0.795
HaarPSI 0.862



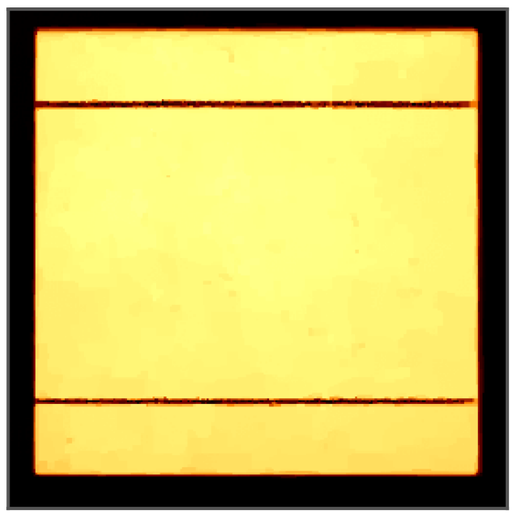
GS-PnP-tuned-5
PSNR 30.66 dB | SSIM 0.622
HaarPSI 0.823



Scalar TV
PSNR 27.78 dB | SSIM 0.675
HaarPSI 0.781



UTV C60
PSNR 31.53 dB | SSIM 0.781
HaarPSI 0.814






## Comparison notes

All methods share the exact frozen measurement and sampling indices.

Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Visual observation: vertical lines remain in zero-fill but are lost by the other methods.